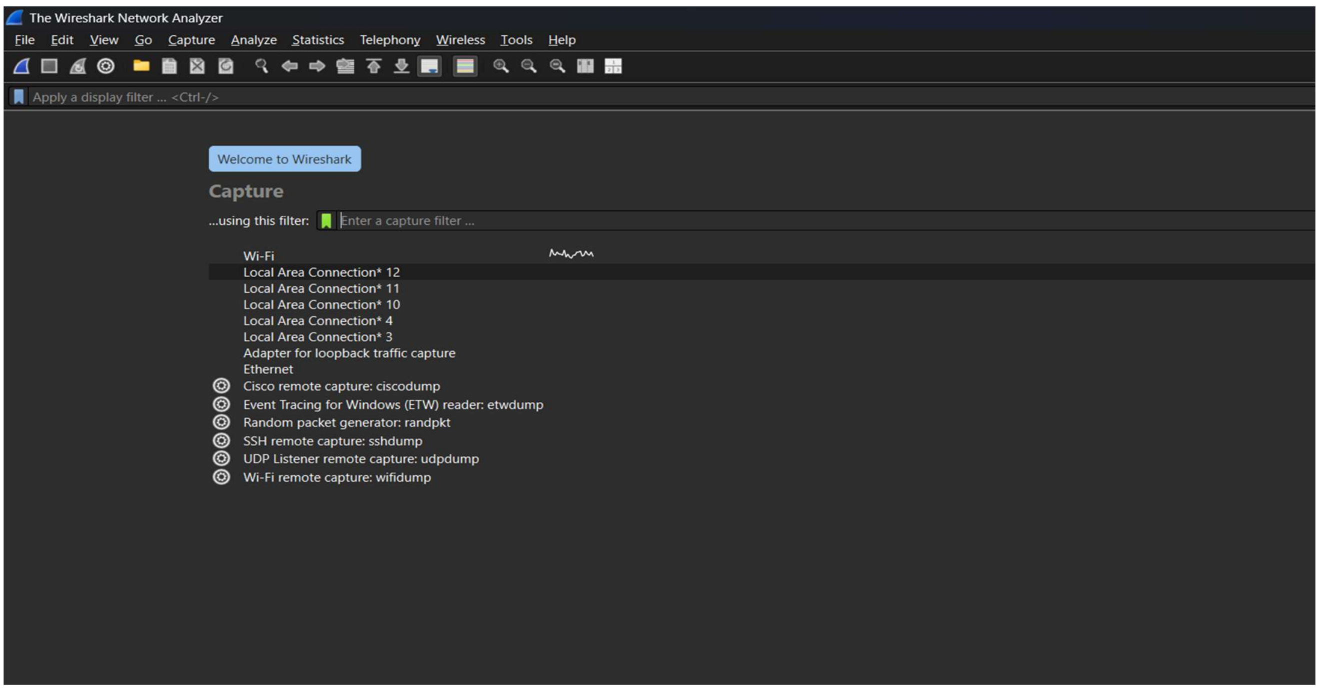
ASSIGNMENT 1

1. **To capture live network packets using Wireshark and identify basic network protocols such as TCP, UDP, IP, and DNS.**

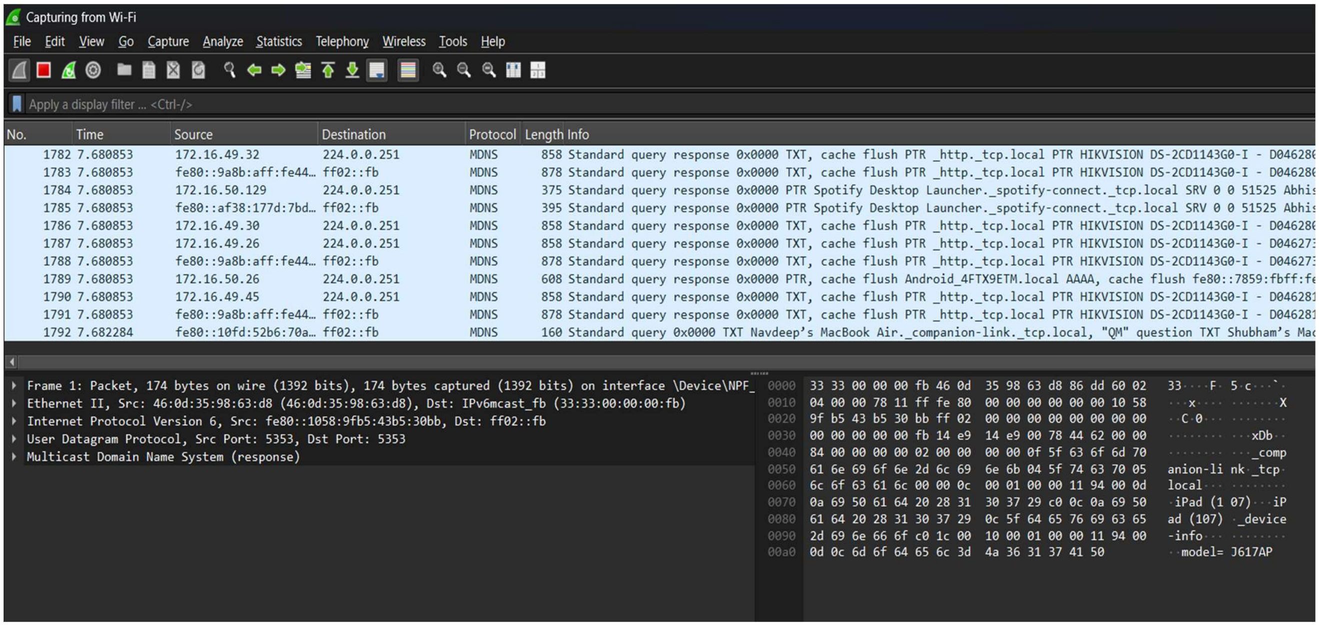
Step 1: Launch Wireshark

- Open **Wireshark**
- You will see a list of **network interfaces**
  - Wi-Fi
  - Ethernet



Step 2: Start Packet Capture

- Double-click on **Wi-Fi** (or active interface)
- Packet capture will start immediately



Step 3: Generate Network Traffic

Perform any of the following:

- Open a web browser
- Visit any website (e.g., google.com)
- Open Command Prompt and run:
- ping google.com

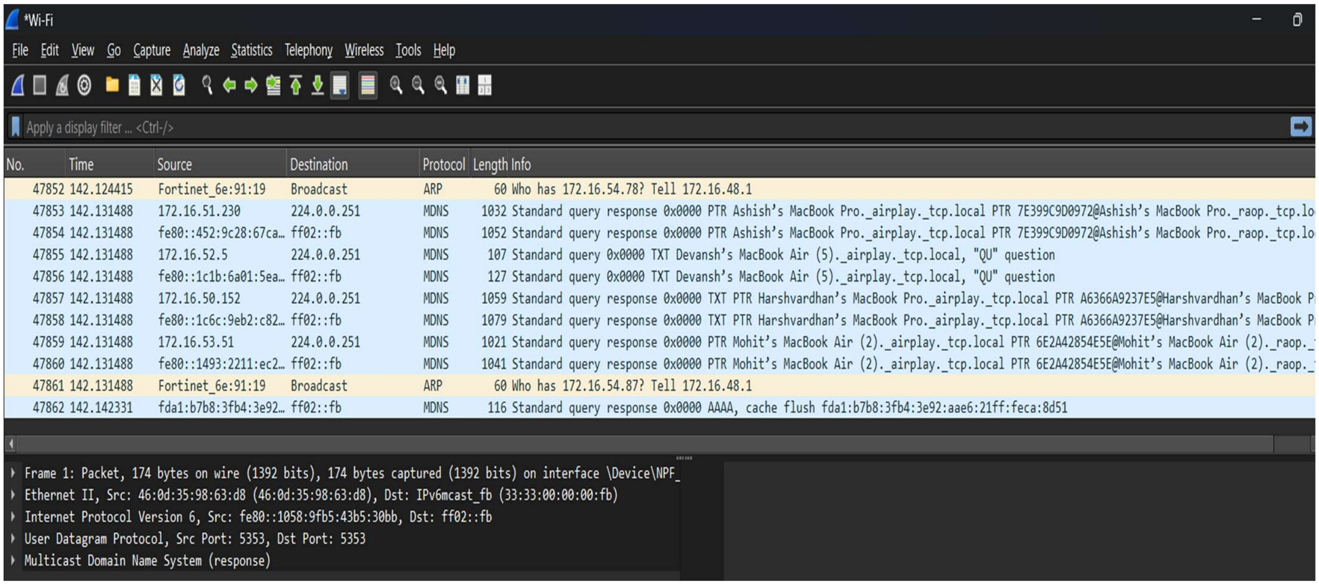
```
C:\Users\dell9>ping google.com

Pinging google.com [142.250.182.238] with 32 bytes of data:
Reply from 142.250.182.238: bytes=32 time=53ms TTL=118
Reply from 142.250.182.238: bytes=32 time=14ms TTL=118
Reply from 142.250.182.238: bytes=32 time=18ms TTL=118
Reply from 142.250.182.238: bytes=32 time=32ms TTL=118

Ping statistics for 142.250.182.238:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 14ms, Maximum = 53ms, Average = 29ms
```

Step 4: Stop Capture

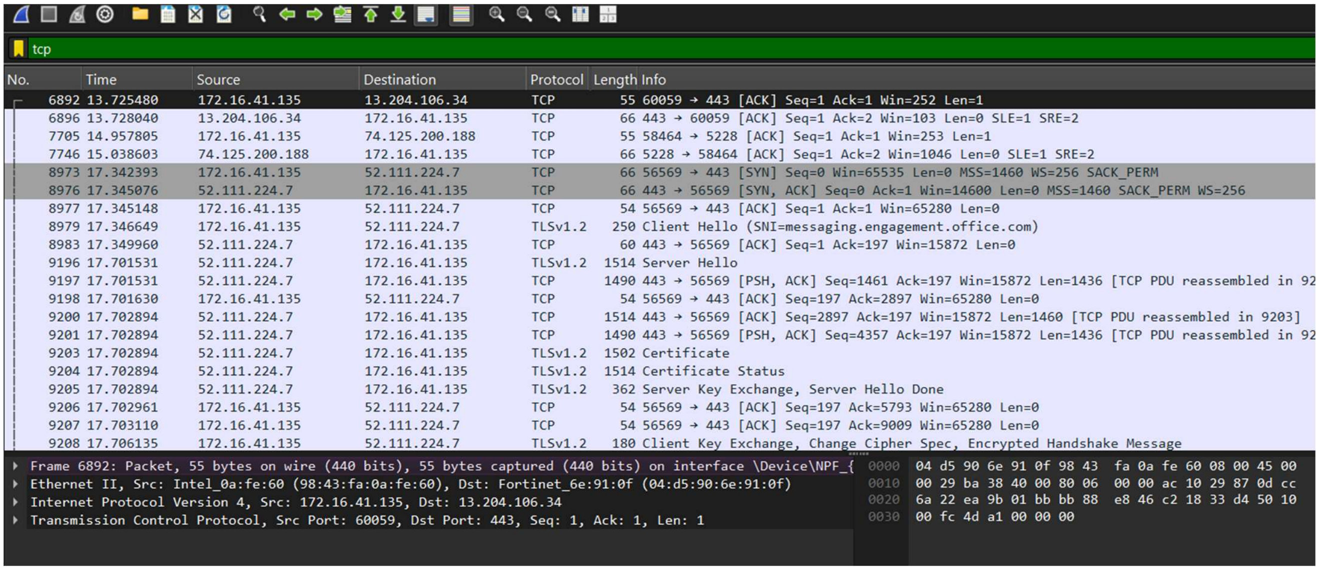
- Click the **red Stop button**
- Captured packets will remain on the screen

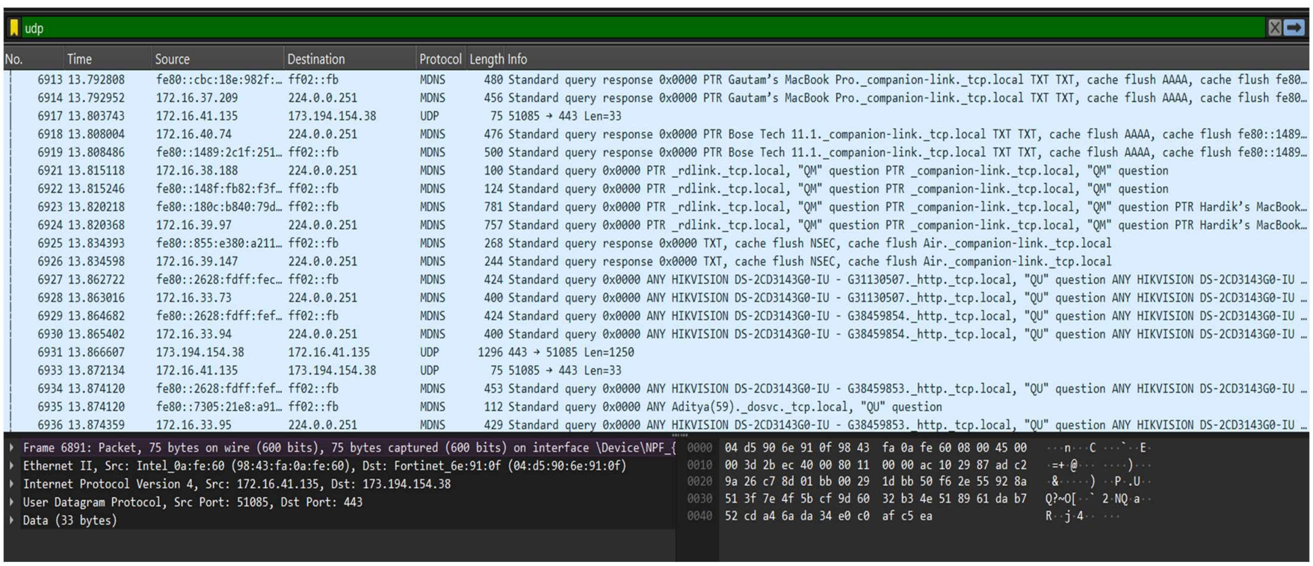


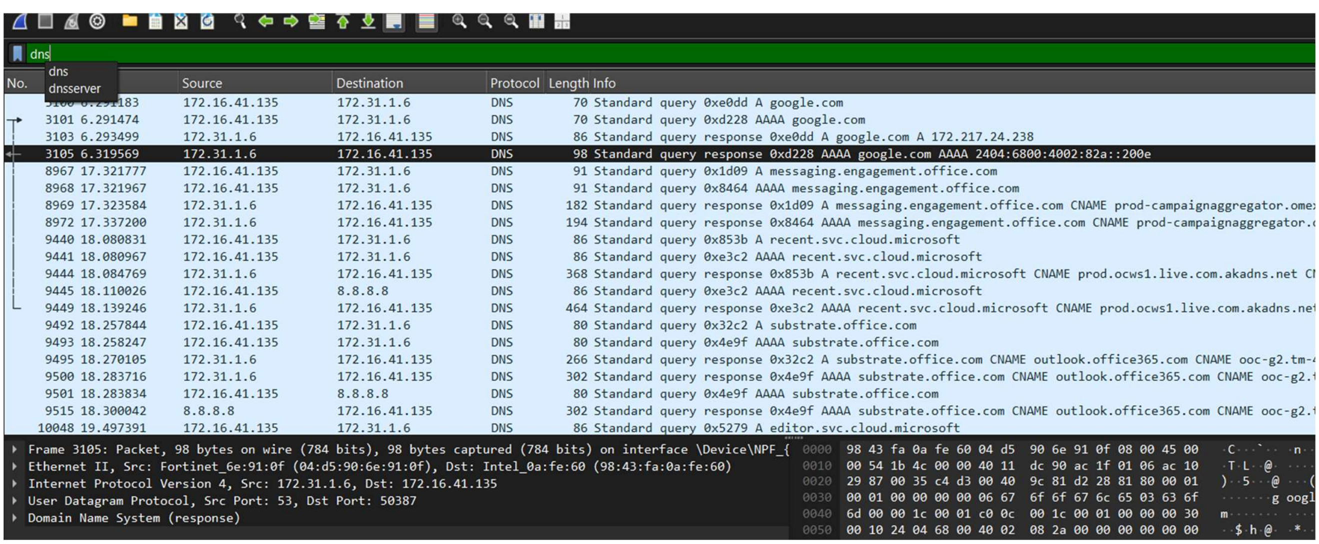
Step 5: Apply Basic Filters

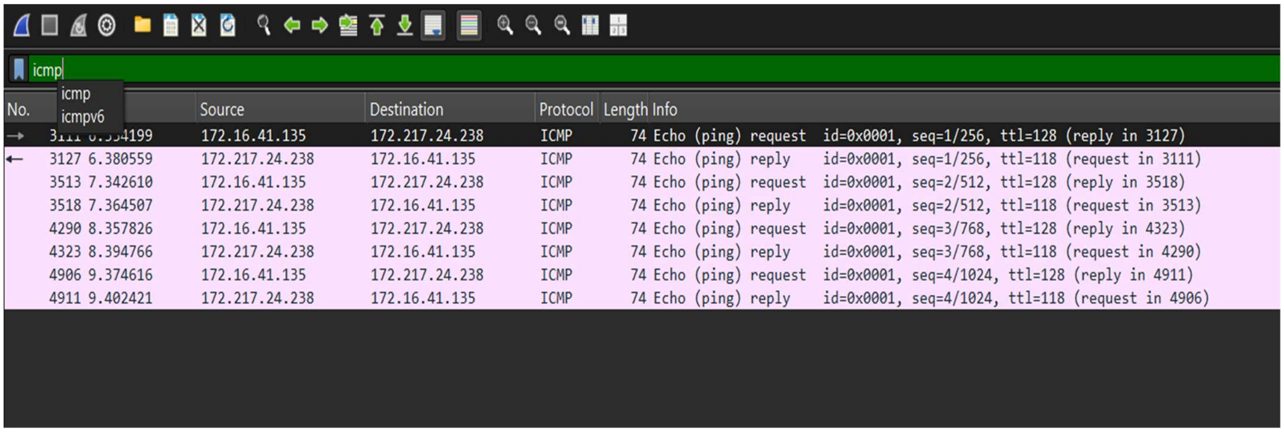
Use the **Display Filter bar**:

Filter	**Purpose**
ip	Show IP packets
tcp	Show TCP packets
udp	Show UDP packets
dns	Show DNS packets
icmp	Show ping packets


tcp
No. Time Source Destination Protocol Length Info
6892 13.725480 172.16.41.135 13.204.106.34 TCP 55 60059 → 443 [ACK] Seq=1 Ack=1 Win=252 Len=1
6896 13.728040 13.204.106.34 172.16.41.135 TCP 66 443 → 60059 [ACK] Seq=1 Ack=2 Win=103 Len=0 SLE=1 SRE=2
7705 14.957805 172.16.41.135 74.125.200.188 TCP 55 58464 → 5228 [ACK] Seq=1 Ack=1 Win=253 Len=1
7746 15.038603 74.125.200.188 172.16.41.135 TCP 66 5228 → 58464 [ACK] Seq=1 Ack=2 Win=1046 Len=0 SLE=1 SRE=2
8973 17.342393 172.16.41.135 52.111.224.7 TCP 66 56569 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
8976 17.345076 52.111.224.7 172.16.41.135 TCP 66 443 → 56569 [SYN, ACK] Seq=0 Ack=1 Win=14600 Len=0 MSS=1460 SACK_PERM WS=256
8977 17.345148 172.16.41.135 52.111.224.7 TCP 54 56569 → 443 [ACK] Seq=1 Ack=1 Win=65280 Len=0
8979 17.346649 172.16.41.135 52.111.224.7 TLSv1.2 250 Client Hello (SNI=messaging.engagement.office.com)
8983 17.349960 52.111.224.7 172.16.41.135 TCP 60 443 → 56569 [ACK] Seq=1 Ack=197 Win=15872 Len=0
9196 17.701531 52.111.224.7 172.16.41.135 TLSv1.2 1514 Server Hello
9197 17.701531 52.111.224.7 172.16.41.135 TCP 1490 443 → 56569 [PSH, ACK] Seq=1461 Ack=197 Win=15872 Len=1436 [TCP PDU reassembled in 92
9198 17.701630 172.16.41.135 52.111.224.7 TCP 54 56569 → 443 [ACK] Seq=197 Ack=2897 Win=65280 Len=0
9200 17.702894 52.111.224.7 172.16.41.135 TCP 1514 443 → 56569 [ACK] Seq=2897 Ack=197 Win=15872 Len=1460 [TCP PDU reassembled in 9203]
9201 17.702894 52.111.224.7 172.16.41.135 TCP 1490 443 → 56569 [PSH, ACK] Seq=4357 Ack=197 Win=15872 Len=1436 [TCP PDU reassembled in 92
9203 17.702894 52.111.224.7 172.16.41.135 TLSv1.2 1502 Certificate
9204 17.702894 52.111.224.7 172.16.41.135 TLSv1.2 1514 Certificate Status
9205 17.702894 52.111.224.7 172.16.41.135 TLSv1.2 362 Server Key Exchange, Server Hello Done
9206 17.702961 172.16.41.135 52.111.224.7 TCP 54 56569 → 443 [ACK] Seq=197 Ack=5793 Win=65280 Len=0
9207 17.703110 172.16.41.135 52.111.224.7 TCP 54 56569 → 443 [ACK] Seq=197 Ack=9009 Win=65280 Len=0
9208 17.706135 172.16.41.135 52.111.224.7 TLSv1.2 180 Client Key Exchange, Change Cipher Spec, Encrypted Handshake Message
Frame 6892: Packet, 55 bytes on wire (440 bits), 55 bytes captured (440 bits) on interface \Device\NPF_{
Ethernet II, Src: Intel_0a:fe:60 (98:43:fa:0a:fe:60), Dst: Fortinet_6e:91:0f (04:d5:90:6e:91:0f)
Internet Protocol Version 4, Src: 172.16.41.135, Dst: 13.204.106.34
Transmission Control Protocol, Src Port: 60059, Dst Port: 443, Seq: 1, Ack: 1, Len: 1
0000 04 d5 90 6e 91 0f 98 43 fa 0a fe 60 08 00 45 00
0010 00 29 ba 38 40 00 80 06 00 00 ac 10 29 87 0d cc
0020 6a 22 ea 9b 01 bb bb 88 e8 46 c2 18 33 d4 50 10
0030 00 fc 4d a1 00 00 00

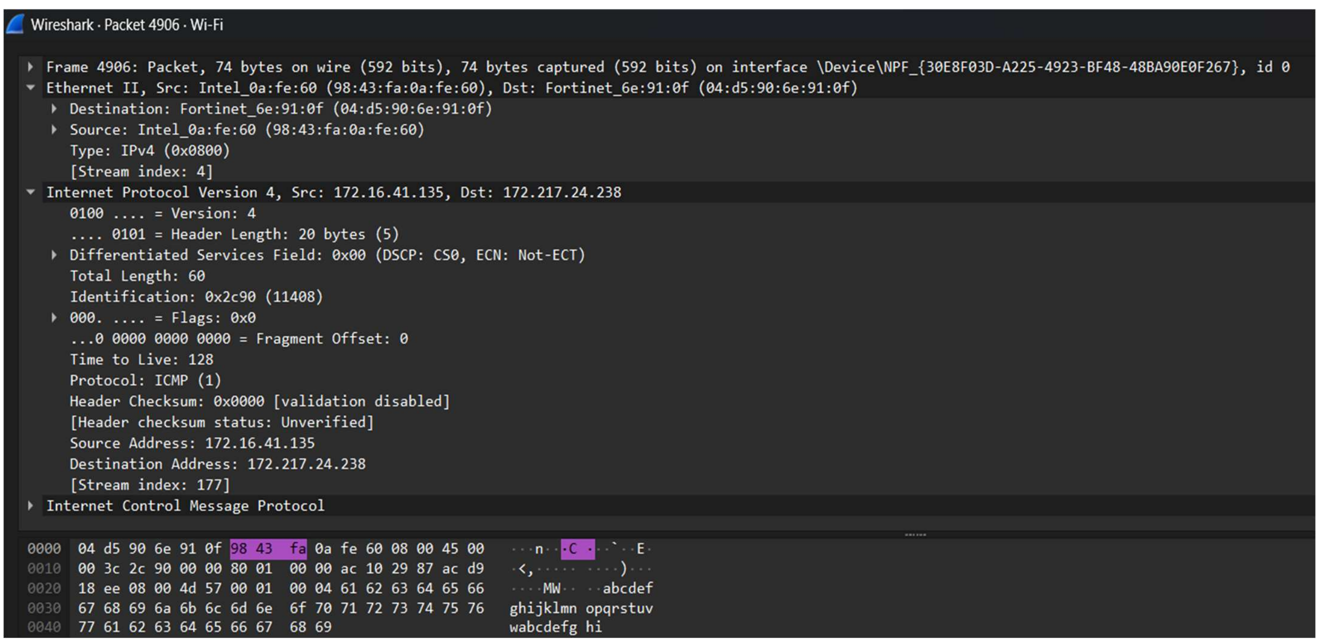

udp
No. Time Source Destination Protocol Length Info
6913 13.792808 fe80::cbc:18e:982f:… ff02::fb MDNS 480 Standard query response 0x0000 PTR Gautam's MacBook Pro._companion-link._tcp.local TXT TXT, cache flush AAAA, cache flush fe80…
6914 13.792952 172.16.37.209 224.0.0.251 MDNS 456 Standard query response 0x0000 PTR Gautam's MacBook Pro._companion-link._tcp.local TXT TXT, cache flush AAAA, cache flush fe80…
6917 13.803743 172.16.41.135 173.194.154.38 UDP 75 51085 → 443 Len=33
6918 13.808004 172.16.40.74 224.0.0.251 MDNS 476 Standard query response 0x0000 PTR Bose Tech 11.1._companion-link._tcp.local TXT TXT, cache flush AAAA, cache flush fe80::1489…
6919 13.808486 fe80::1489:2c1f:251… ff02::fb MDNS 500 Standard query response 0x0000 PTR Bose Tech 11.1._companion-link._tcp.local TXT TXT, cache flush AAAA, cache flush fe80::1489…
6921 13.815118 172.16.38.188 224.0.0.251 MDNS 100 Standard query 0x0000 PTR _rdlink._tcp.local, "QM" question PTR _companion-link._tcp.local, "QM" question
6922 13.815246 fe80::148f:fb82:f3f… ff02::fb MDNS 124 Standard query 0x0000 PTR _rdlink._tcp.local, "QM" question PTR _companion-link._tcp.local, "QM" question
6923 13.820218 fe80::180c:b840:79d… ff02::fb MDNS 781 Standard query 0x0000 PTR _rdlink._tcp.local, "QM" question PTR _companion-link._tcp.local, "QM" question PTR Hardik's MacBook…
6924 13.820368 172.16.39.97 224.0.0.251 MDNS 757 Standard query 0x0000 PTR _rdlink._tcp.local, "QM" question PTR _companion-link._tcp.local, "QM" question PTR Hardik's MacBook…
6925 13.834393 fe80::855:e380:a211… ff02::fb MDNS 268 Standard query response 0x0000 TXT, cache flush NSEC, cache flush Air._companion-link._tcp.local
6926 13.834598 172.16.39.147 224.0.0.251 MDNS 244 Standard query response 0x0000 TXT, cache flush NSEC, cache flush Air._companion-link._tcp.local
6927 13.862722 fe80::2628:fdff:fec… ff02::fb MDNS 424 Standard query 0x0000 ANY HIKVISION DS-2CD3143G0-IU - G31130507._http._tcp.local, "QU" question ANY HIKVISION DS-2CD3143G0-IU …
6928 13.863016 172.16.33.73 224.0.0.251 MDNS 400 Standard query 0x0000 ANY HIKVISION DS-2CD3143G0-IU - G31130507._http._tcp.local, "QU" question ANY HIKVISION DS-2CD3143G0-IU …
6929 13.864682 fe80::2628:fdff:fef… ff02::fb MDNS 424 Standard query 0x0000 ANY HIKVISION DS-2CD3143G0-IU - G38459854._http._tcp.local, "QU" question ANY HIKVISION DS-2CD3143G0-IU …
6930 13.865402 172.16.33.94 224.0.0.251 MDNS 400 Standard query 0x0000 ANY HIKVISION DS-2CD3143G0-IU - G38459854._http._tcp.local, "QU" question ANY HIKVISION DS-2CD3143G0-IU …
6931 13.866607 173.194.154.38 172.16.41.135 UDP 1296 443 → 51085 Len=1250
6933 13.872134 172.16.41.135 173.194.154.38 UDP 75 51085 → 443 Len=33
6934 13.874120 fe80::2628:fdff:fef… ff02::fb MDNS 453 Standard query 0x0000 ANY HIKVISION DS-2CD3143G0-IU - G38459853._http._tcp.local, "QU" question ANY HIKVISION DS-2CD3143G0-IU …
6935 13.874120 fe80::7305:21e8:a91… ff02::fb MDNS 112 Standard query 0x0000 ANY Aditya(59)._dosvc._tcp.local, "QU" question
6936 13.874359 172.16.33.95 224.0.0.251 MDNS 429 Standard query 0x0000 ANY HIKVISION DS-2CD3143G0-IU - G38459853._http._tcp.local, "QU" question ANY HIKVISION DS-2CD3143G0-IU …
Frame 6891: Packet, 75 bytes on wire (600 bits), 75 bytes captured (600 bits) on interface \Device\NPF_{
Ethernet II, Src: Intel_0a:fe:60 (98:43:fa:0a:fe:60), Dst: Fortinet_6e:91:0f (04:d5:90:6e:91:0f)
Internet Protocol Version 4, Src: 172.16.41.135, Dst: 173.194.154.38
User Datagram Protocol, Src Port: 51085, Dst Port: 443
Data (33 bytes)
0000 04 d5 90 6e 91 0f 98 43 fa 0a fe 60 08 00 45 00
0010 00 3d 2b ec 40 00 80 11 00 00 ac 10 29 87 ad c2
0020 9a 26 c7 8d 01 bb 00 29 1d bb 50 f6 2e 55 92 8a
0030 51 3f 7e 4f 5b cf 9d 60 32 b3 4e 51 89 61 da b7
0040 52 cd a4 6a da 34 e0 c0 af c5 ea


dns
dns
dnsserver
No. Source Destination Protocol Length Info
172.16.41.135 172.31.1.6 DNS 70 Standard query 0xe0dd A google.com
3101 6.291474 172.16.41.135 172.31.1.6 DNS 70 Standard query 0xd228 AAAA google.com
3103 6.293499 172.31.1.6 172.16.41.135 DNS 86 Standard query response 0xe0dd A google.com A 172.217.24.238
3105 6.319569 172.31.1.6 172.16.41.135 DNS 98 Standard query response 0xd228 AAAA google.com AAAA 2404:6800:4002:82a::200e
8967 17.321777 172.16.41.135 172.31.1.6 DNS 91 Standard query 0x1d09 A messaging.engagement.office.com
8968 17.321967 172.16.41.135 172.31.1.6 DNS 91 Standard query 0x8464 AAAA messaging.engagement.office.com
8969 17.323584 172.31.1.6 172.16.41.135 DNS 182 Standard query response 0x1d09 A messaging.engagement.office.com CNAME prod-campaignaggregator.ome
8972 17.337200 172.31.1.6 172.16.41.135 DNS 194 Standard query response 0x8464 AAAA messaging.engagement.office.com CNAME prod-campaignaggregator.
9440 18.080831 172.16.41.135 172.31.1.6 DNS 86 Standard query 0x853b A recent.svc.cloud.microsoft
9441 18.080967 172.16.41.135 172.31.1.6 DNS 86 Standard query 0xe3c2 AAAA recent.svc.cloud.microsoft
9444 18.084769 172.31.1.6 172.16.41.135 DNS 368 Standard query response 0x853b A recent.svc.cloud.microsoft CNAME prod.ocws1.live.com.akadns.net C
9445 18.110026 172.16.41.135 8.8.8.8 DNS 86 Standard query 0xe3c2 AAAA recent.svc.cloud.microsoft
9449 18.139246 172.31.1.6 172.16.41.135 DNS 464 Standard query response 0xe3c2 AAAA recent.svc.cloud.microsoft CNAME prod.ocws1.live.com.akadns.ne
9492 18.257844 172.16.41.135 172.31.1.6 DNS 80 Standard query 0x32c2 A substrate.office.com
9493 18.258247 172.16.41.135 172.31.1.6 DNS 80 Standard query 0x4e9f AAAA substrate.office.com
9495 18.270105 172.31.1.6 172.16.41.135 DNS 266 Standard query response 0x32c2 A substrate.office.com CNAME outlook.office365.com CNAME ooc-g2.tm-
9500 18.283716 172.31.1.6 172.16.41.135 DNS 302 Standard query response 0x4e9f AAAA substrate.office.com CNAME outlook.office365.com CNAME ooc-g2.
9501 18.283834 172.16.41.135 8.8.8.8 DNS 80 Standard query 0x4e9f AAAA substrate.office.com
9515 18.300042 8.8.8.8 172.16.41.135 DNS 302 Standard query response 0x4e9f AAAA substrate.office.com CNAME outlook.office365.com CNAME ooc-g2.
10048 19.497391 172.16.41.135 172.31.1.6 DNS 86 Standard query 0x5279 A editor.svc.cloud.microsoft
Frame 3105: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface \Device\NPF_{
Ethernet II, Src: Fortinet_6e:91:0f (04:d5:90:6e:91:0f), Dst: Intel_0a:fe:60 (98:43:fa:0a:fe:60)
Internet Protocol Version 4, Src: 172.31.1.6, Dst: 172.16.41.135
User Datagram Protocol, Src Port: 53, Dst Port: 50387
Domain Name System (response)
0000 98 43 fa 0a fe 60 04 d5 90 6e 91 0f 08 00 45 00
0010 00 54 1b 4c 00 00 40 11 dc 90 ac 1f 01 06 ac 10
0020 29 87 00 35 c4 d3 00 40 9c 81 d2 28 81 80 00 01
0030 00 01 00 00 00 00 06 67 6f 6f 67 6c 65 03 63 6f
0040 6d 00 00 1c 00 01 c0 0c 00 1c 00 01 00 00 00 30
0050 00 10 24 04 68 00 40 02 08 2a 00 00 00 00 00 00



Step 6: Analyze a Packet

- Click on any packet
- Observe:
  - **Ethernet header**
  - **IP header**
  - **TCP/UDP header**
  - Payload data



Observation

Interface used:

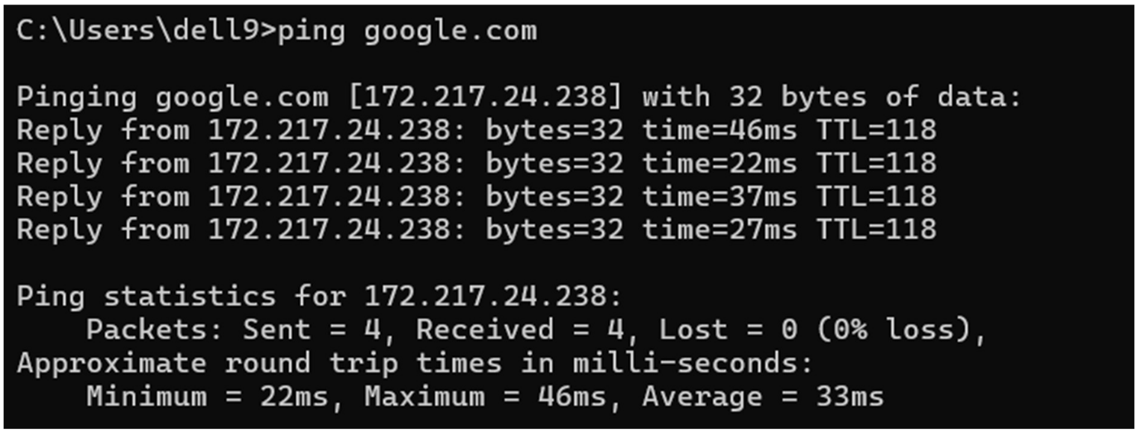
- Protocols observed:
- Source and destination IP addresses
- Packet length

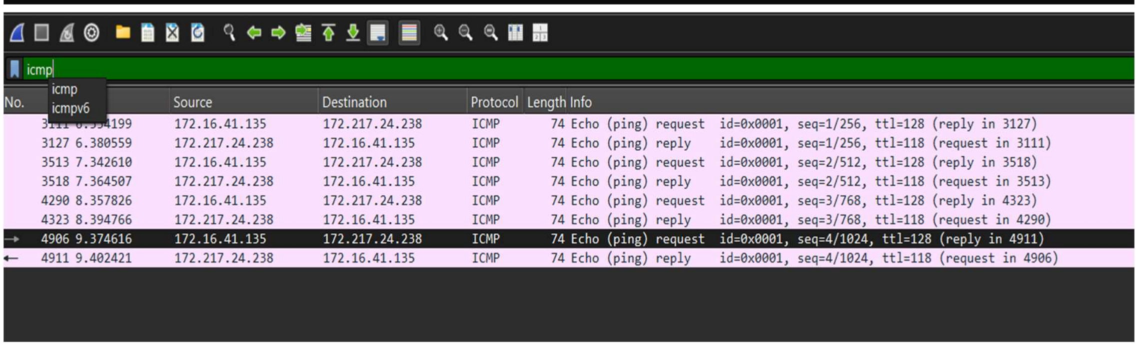
Result

Live network packets were successfully captured using Wireshark, and basic network protocols were identified and analyzed.

ASSIGNMENT 2

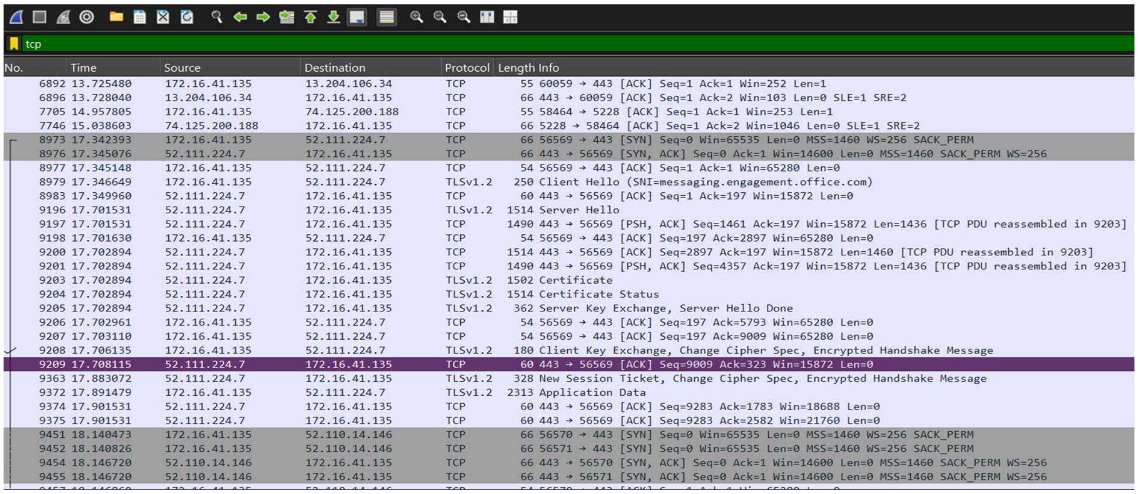
1. To capture and analyze ICMP packets (Ping request and reply)

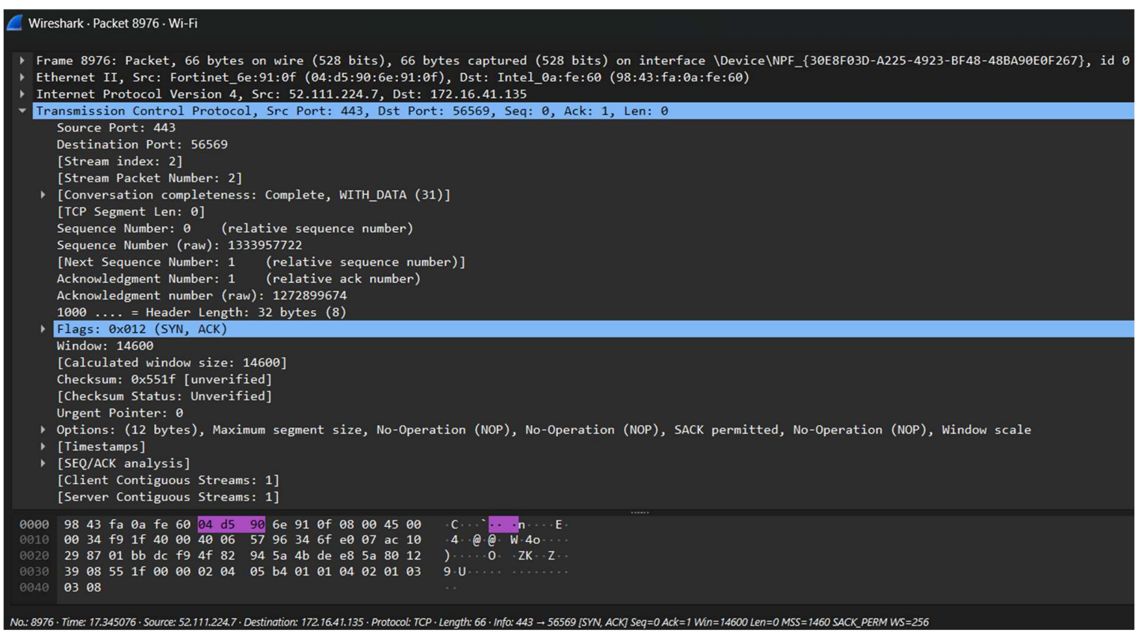




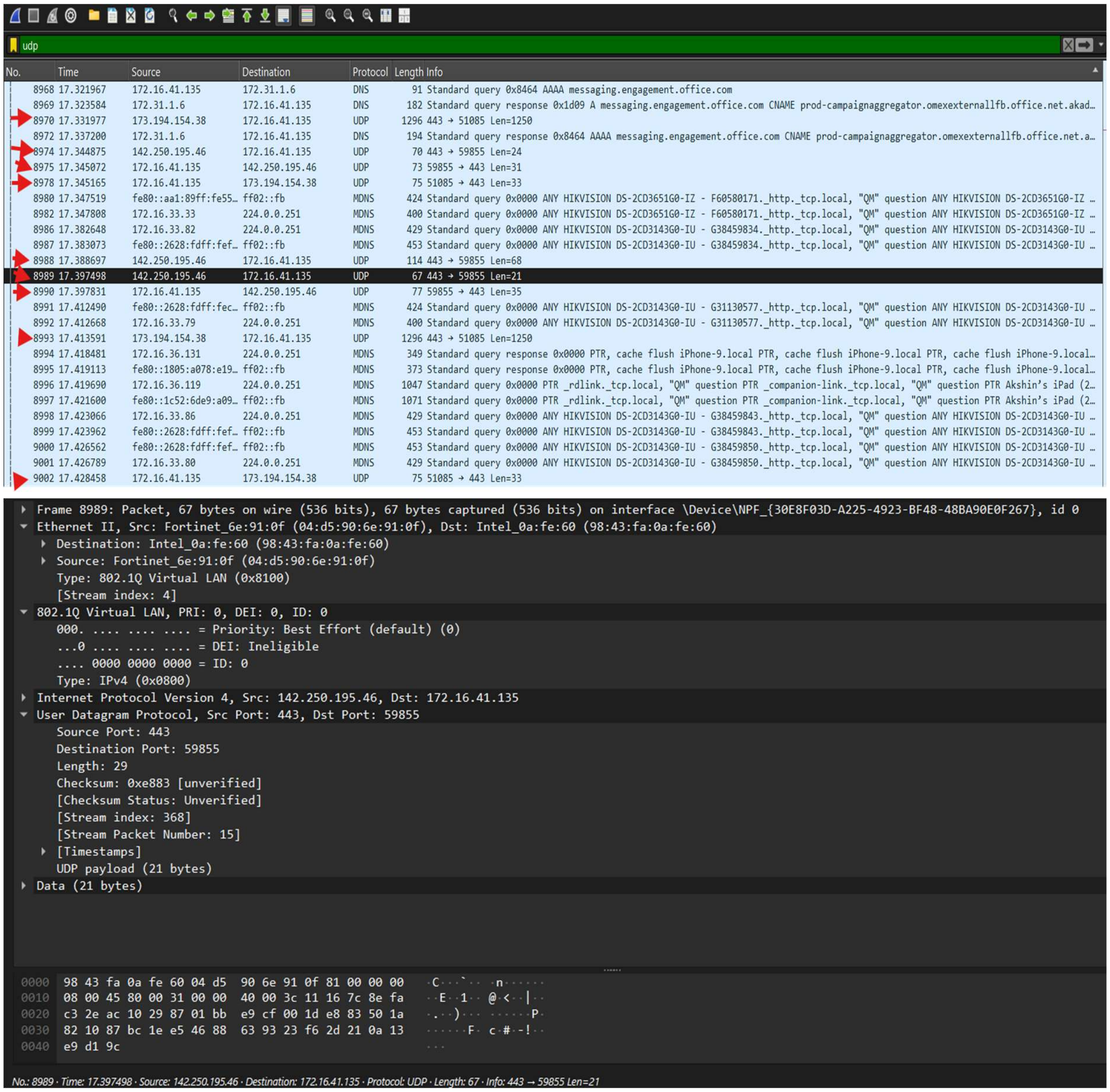
No.		Source	Destination	Protocol	Length	Info
3111	[illegible]4199	172.16.41.135	172.217.24.238	ICMP	74	Echo (ping) request id=0x0001, seq=1/256, ttl=128 (reply in 3127)
3127	6.380559	172.217.24.238	172.16.41.135	ICMP	74	Echo (ping) reply id=0x0001, seq=1/256, ttl=118 (request in 3111)
3513	7.342610	172.16.41.135	172.217.24.238	ICMP	74	Echo (ping) request id=0x0001, seq=2/512, ttl=128 (reply in 3518)
3518	7.364507	172.217.24.238	172.16.41.135	ICMP	74	Echo (ping) reply id=0x0001, seq=2/512, ttl=118 (request in 3513)
4290	8.357826	172.16.41.135	172.217.24.238	ICMP	74	Echo (ping) request id=0x0001, seq=3/768, ttl=128 (reply in 4323)
4323	8.394766	172.217.24.238	172.16.41.135	ICMP	74	Echo (ping) reply id=0x0001, seq=3/768, ttl=118 (request in 4290)
4906	9.374616	172.16.41.135	172.217.24.238	ICMP	74	Echo (ping) request id=0x0001, seq=4/1024, ttl=128 (reply in 4911)
4911	9.402421	172.217.24.238	172.16.41.135	ICMP	74	Echo (ping) reply id=0x0001, seq=4/1024, ttl=118 (request in 4906)

2. To analyze TCP packets and observe the TCP three-way handshake

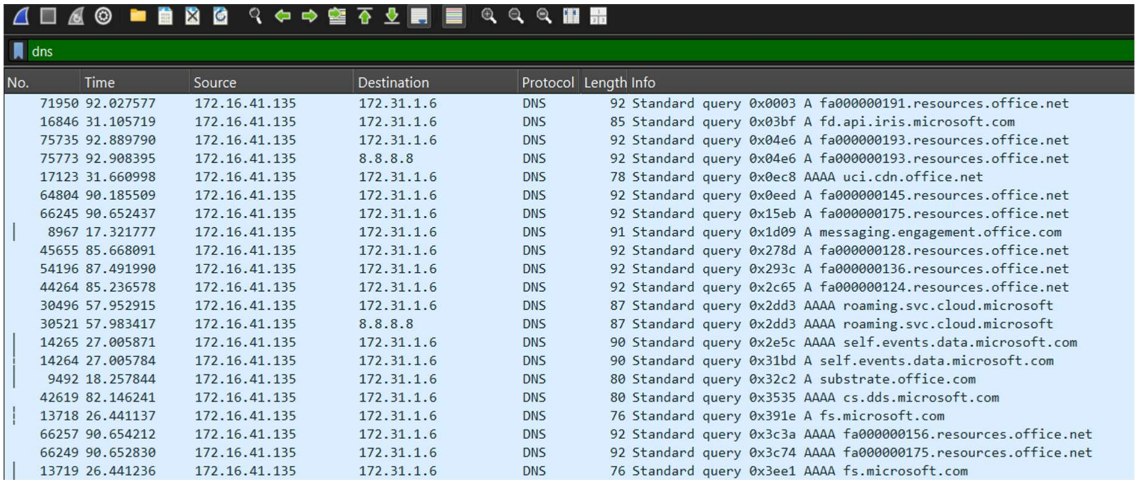




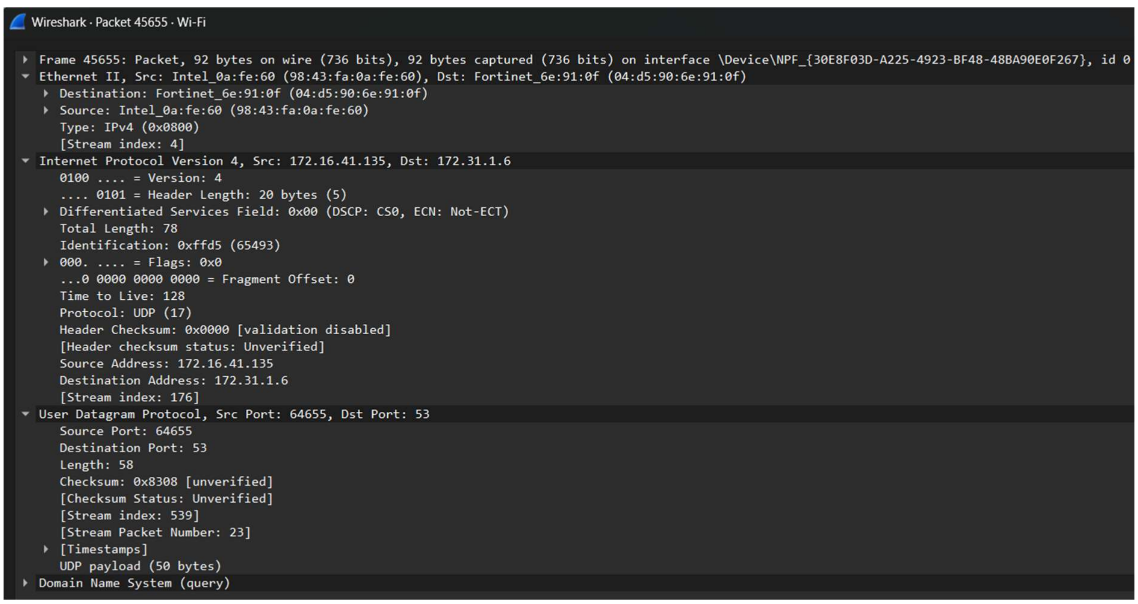
3. To capture and analyze UDP packets



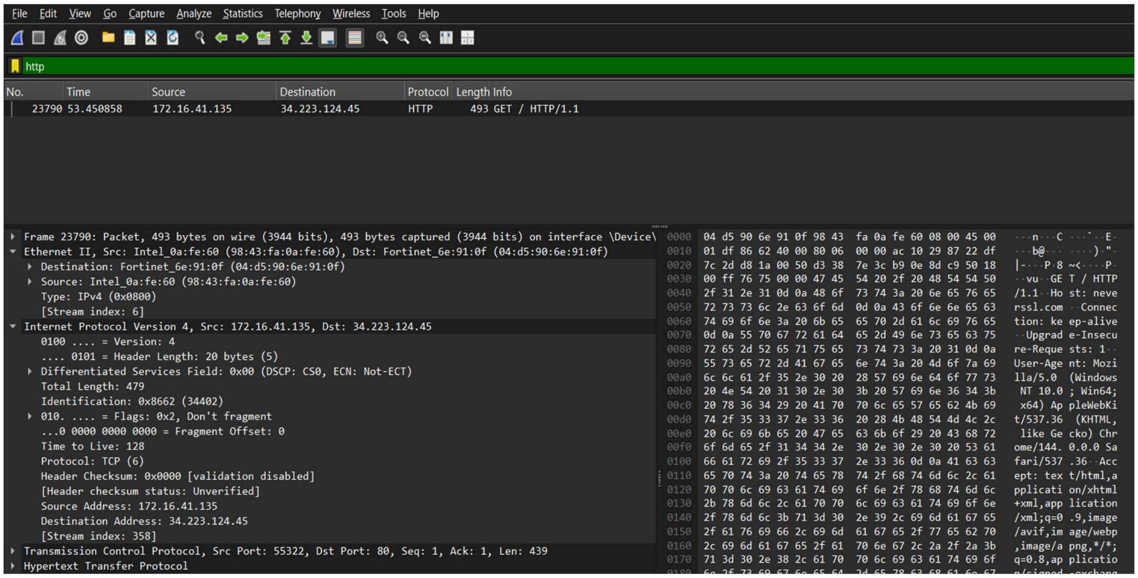
4. To capture and analyze DNS query and response packets



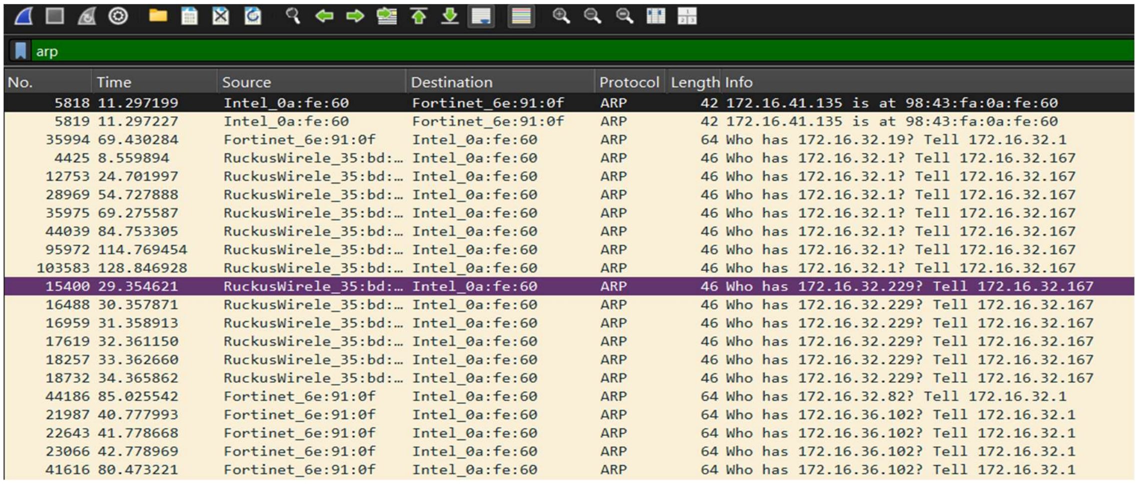
No.	Time	Source	Destination	Protocol	Length	Info
71950	92.027577	172.16.41.135	172.31.1.6	DNS	92	Standard query 0x0003 A fa000000191.resources.office.net
16846	31.105719	172.16.41.135	172.31.1.6	DNS	85	Standard query 0x03bf A fd.api.iris.microsoft.com
75735	92.889790	172.16.41.135	172.31.1.6	DNS	92	Standard query 0x04e6 A fa000000193.resources.office.net
75773	92.908395	172.16.41.135	8.8.8.8	DNS	92	Standard query 0x04e6 A fa000000193.resources.office.net
17123	31.660998	172.16.41.135	172.31.1.6	DNS	78	Standard query 0x0ec8 AAAA uci.cdn.office.net
64804	90.185509	172.16.41.135	172.31.1.6	DNS	92	Standard query 0x0eed A fa000000145.resources.office.net
66245	90.652437	172.16.41.135	172.31.1.6	DNS	92	Standard query 0x15eb A fa000000175.resources.office.net
8967	17.321777	172.16.41.135	172.31.1.6	DNS	91	Standard query 0x1d09 A messaging.engagement.office.com
45655	85.668091	172.16.41.135	172.31.1.6	DNS	92	Standard query 0x278d A fa000000128.resources.office.net
54196	87.491990	172.16.41.135	172.31.1.6	DNS	92	Standard query 0x293c A fa000000136.resources.office.net
44264	85.236578	172.16.41.135	172.31.1.6	DNS	92	Standard query 0x2c65 A fa000000124.resources.office.net
30496	57.952915	172.16.41.135	172.31.1.6	DNS	87	Standard query 0x2dd3 AAAA roaming.svc.cloud.microsoft
30521	57.983417	172.16.41.135	8.8.8.8	DNS	87	Standard query 0x2dd3 AAAA roaming.svc.cloud.microsoft
14265	27.005871	172.16.41.135	172.31.1.6	DNS	90	Standard query 0x2e5c AAAA self.events.data.microsoft.com
14264	27.005784	172.16.41.135	172.31.1.6	DNS	90	Standard query 0x31bd A self.events.data.microsoft.com
9492	18.257844	172.16.41.135	172.31.1.6	DNS	80	Standard query 0x32c2 A substrate.office.com
42619	82.146241	172.16.41.135	172.31.1.6	DNS	80	Standard query 0x3535 AAAA cs.dds.microsoft.com
13718	26.441137	172.16.41.135	172.31.1.6	DNS	76	Standard query 0x391e A fs.microsoft.com
66257	90.654212	172.16.41.135	172.31.1.6	DNS	92	Standard query 0x3c3a AAAA fa000000156.resources.office.net
66249	90.652830	172.16.41.135	172.31.1.6	DNS	92	Standard query 0x3c74 AAAA fa000000175.resources.office.net
13719	26.441236	172.16.41.135	172.31.1.6	DNS	76	Standard query 0x3ee1 AAAA fs.microsoft.com



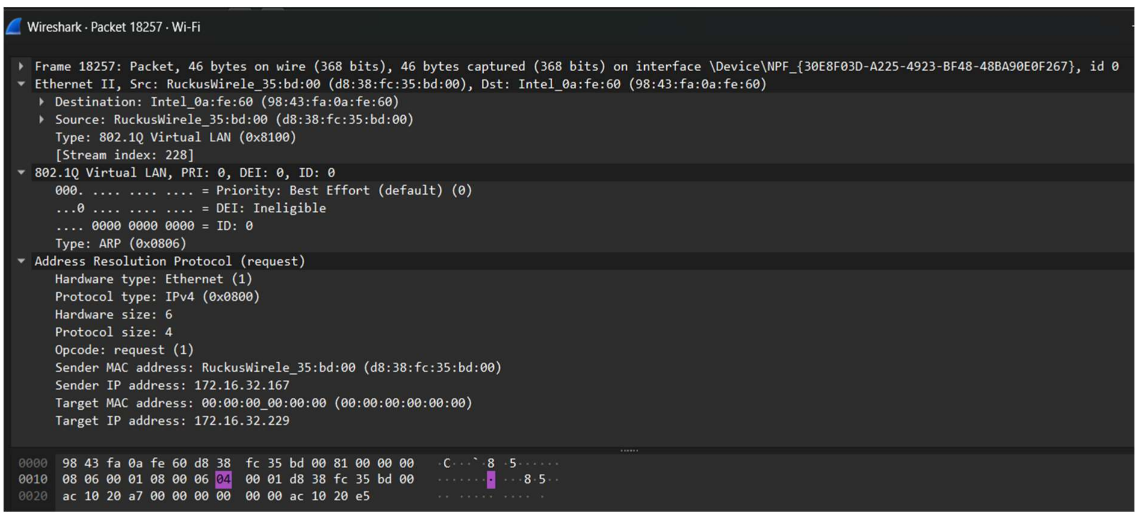
5. To capture and analyze HTTP packets



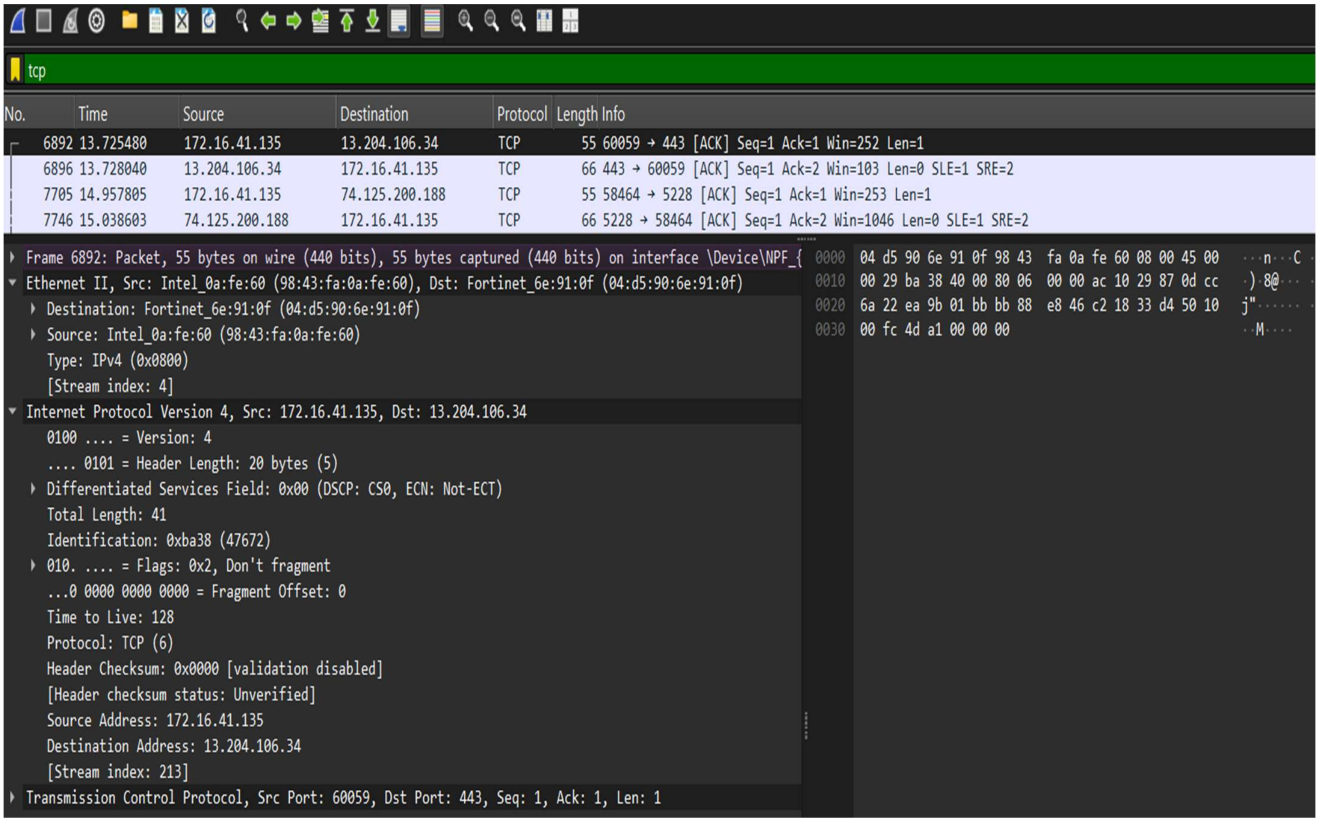
6. To analyze ARP request and reply packets



No.	Time	Source	Destination	Protocol	Length	Info
5818	11.297199	Intel_0a:fe:60	Fortinet_6e:91:0f	ARP	42	172.16.41.135 is at 98:43:fa:0a:fe:60
5819	11.297227	Intel_0a:fe:60	Fortinet_6e:91:0f	ARP	42	172.16.41.135 is at 98:43:fa:0a:fe:60
35994	69.430284	Fortinet_6e:91:0f	Intel_0a:fe:60	ARP	64	Who has 172.16.32.19? Tell 172.16.32.1
4425	8.559894	RuckusWirele_35:bd:…	Intel_0a:fe:60	ARP	46	Who has 172.16.32.1? Tell 172.16.32.167
12753	24.701997	RuckusWirele_35:bd:…	Intel_0a:fe:60	ARP	46	Who has 172.16.32.1? Tell 172.16.32.167
28969	54.727888	RuckusWirele_35:bd:…	Intel_0a:fe:60	ARP	46	Who has 172.16.32.1? Tell 172.16.32.167
35975	69.275587	RuckusWirele_35:bd:…	Intel_0a:fe:60	ARP	46	Who has 172.16.32.1? Tell 172.16.32.167
44039	84.753305	RuckusWirele_35:bd:…	Intel_0a:fe:60	ARP	46	Who has 172.16.32.1? Tell 172.16.32.167
95972	114.769454	RuckusWirele_35:bd:…	Intel_0a:fe:60	ARP	46	Who has 172.16.32.1? Tell 172.16.32.167
103583	128.846928	RuckusWirele_35:bd:…	Intel_0a:fe:60	ARP	46	Who has 172.16.32.1? Tell 172.16.32.167
15400	29.354621	RuckusWirele_35:bd:…	Intel_0a:fe:60	ARP	46	Who has 172.16.32.229? Tell 172.16.32.167
16488	30.357871	RuckusWirele_35:bd:…	Intel_0a:fe:60	ARP	46	Who has 172.16.32.229? Tell 172.16.32.167
16959	31.358913	RuckusWirele_35:bd:…	Intel_0a:fe:60	ARP	46	Who has 172.16.32.229? Tell 172.16.32.167
17619	32.361150	RuckusWirele_35:bd:…	Intel_0a:fe:60	ARP	46	Who has 172.16.32.229? Tell 172.16.32.167
18257	33.362660	RuckusWirele_35:bd:…	Intel_0a:fe:60	ARP	46	Who has 172.16.32.229? Tell 172.16.32.167
18732	34.365862	RuckusWirele_35:bd:…	Intel_0a:fe:60	ARP	46	Who has 172.16.32.229? Tell 172.16.32.167
44186	85.025542	Fortinet_6e:91:0f	Intel_0a:fe:60	ARP	64	Who has 172.16.32.82? Tell 172.16.32.1
21987	40.777993	Fortinet_6e:91:0f	Intel_0a:fe:60	ARP	64	Who has 172.16.36.102? Tell 172.16.32.1
22643	41.778668	Fortinet_6e:91:0f	Intel_0a:fe:60	ARP	64	Who has 172.16.36.102? Tell 172.16.32.1
23066	42.778969	Fortinet_6e:91:0f	Intel_0a:fe:60	ARP	64	Who has 172.16.36.102? Tell 172.16.32.1
41616	80.473221	Fortinet_6e:91:0f	Intel_0a:fe:60	ARP	64	Who has 172.16.36.102? Tell 172.16.32.1

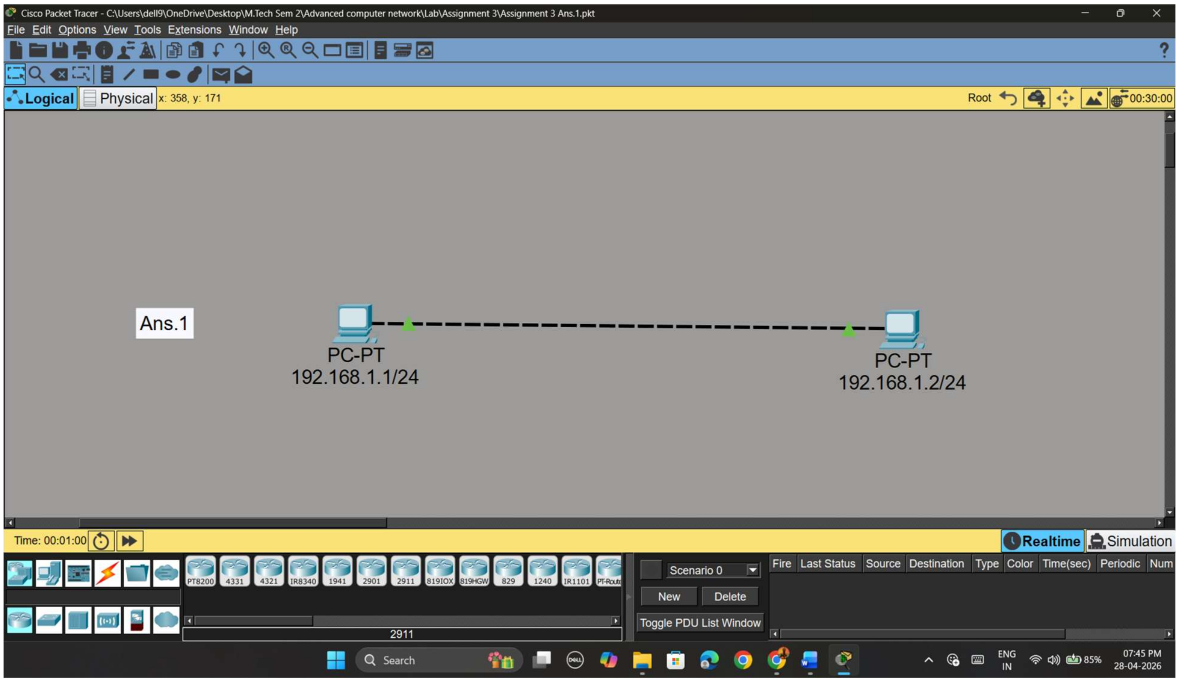


7. **To identify source and destination MAC and IP addresses in captured packets.**

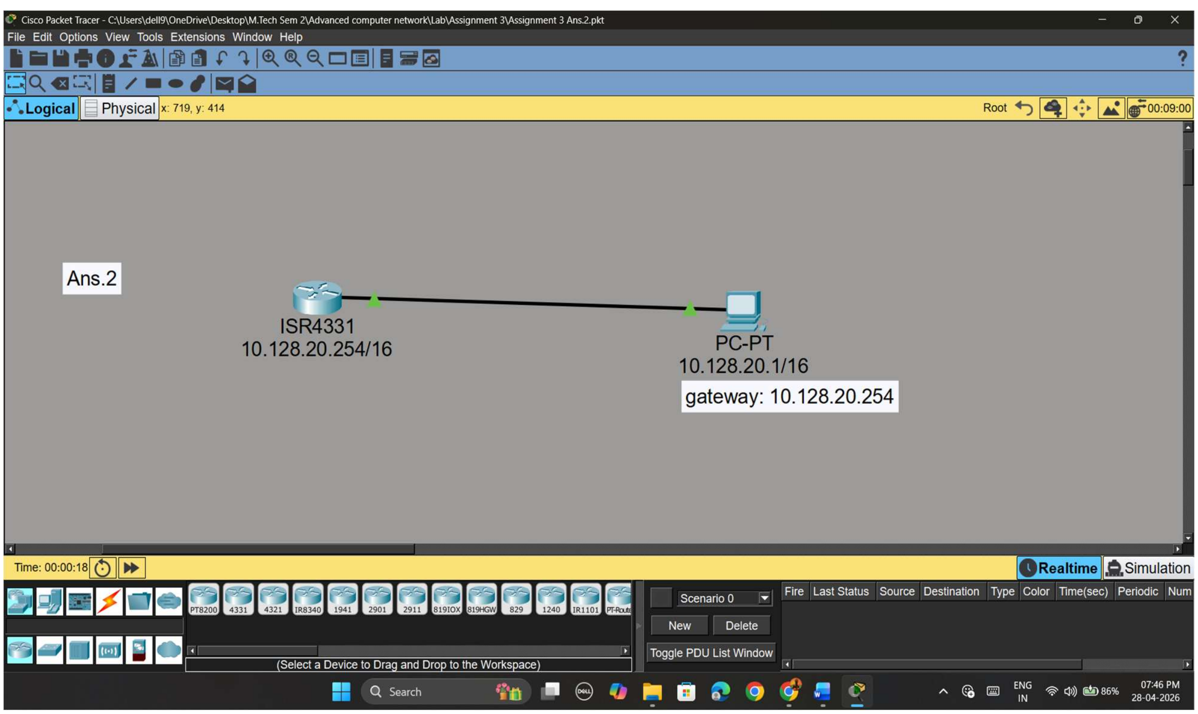


ASSIGNMENT 3

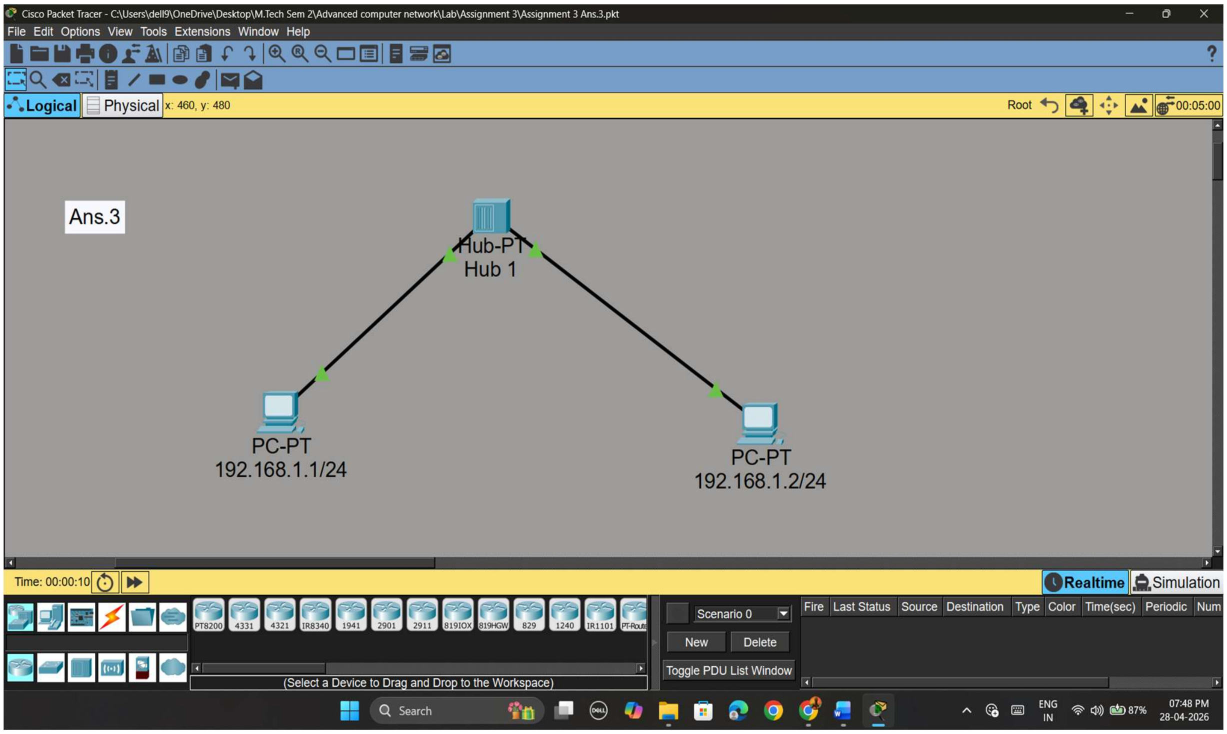
Ans.1



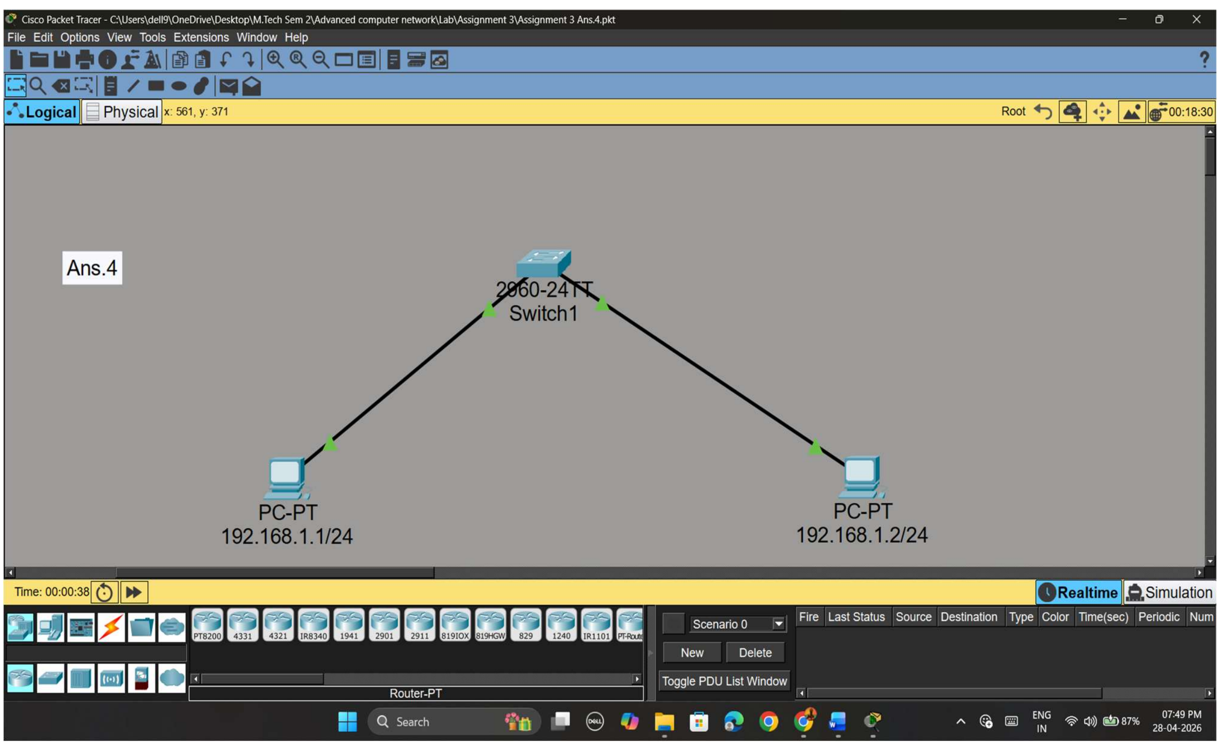
Ans.2



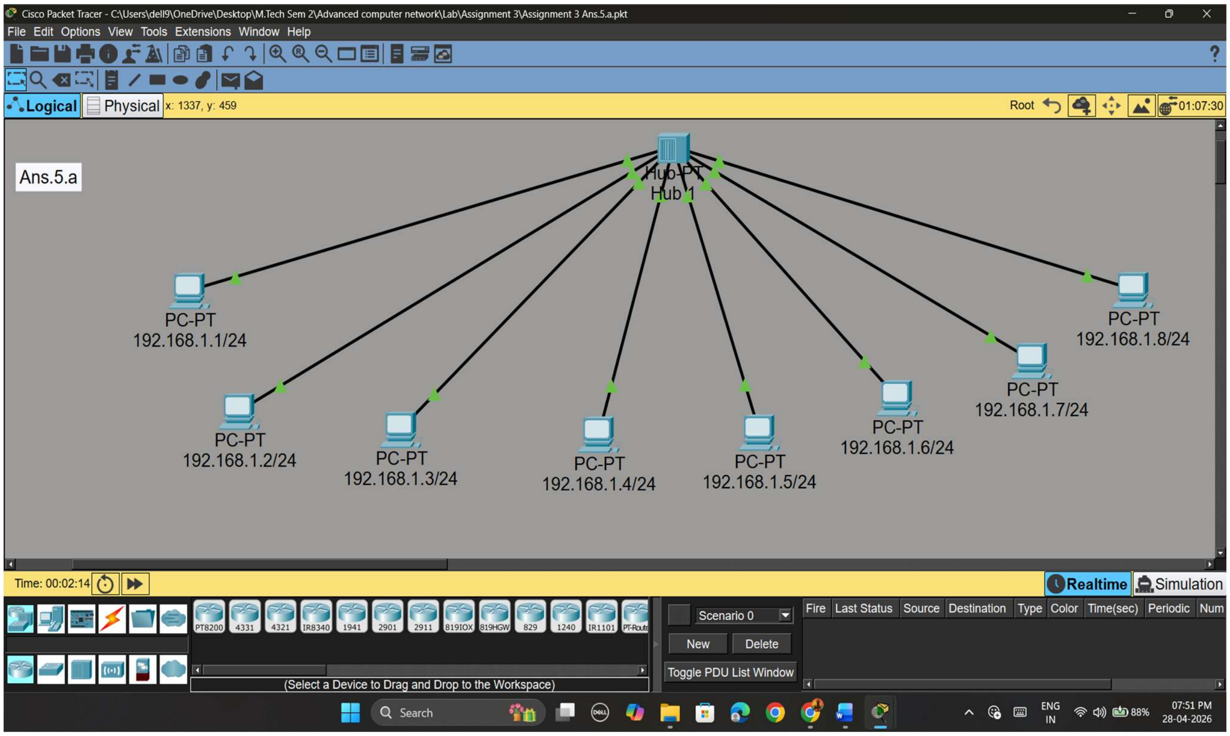
Ans.3


Ans.3
Hub-PT
Hub 1
PC-PT
192.168.1.1/24
PC-PT
192.168.1.2/24

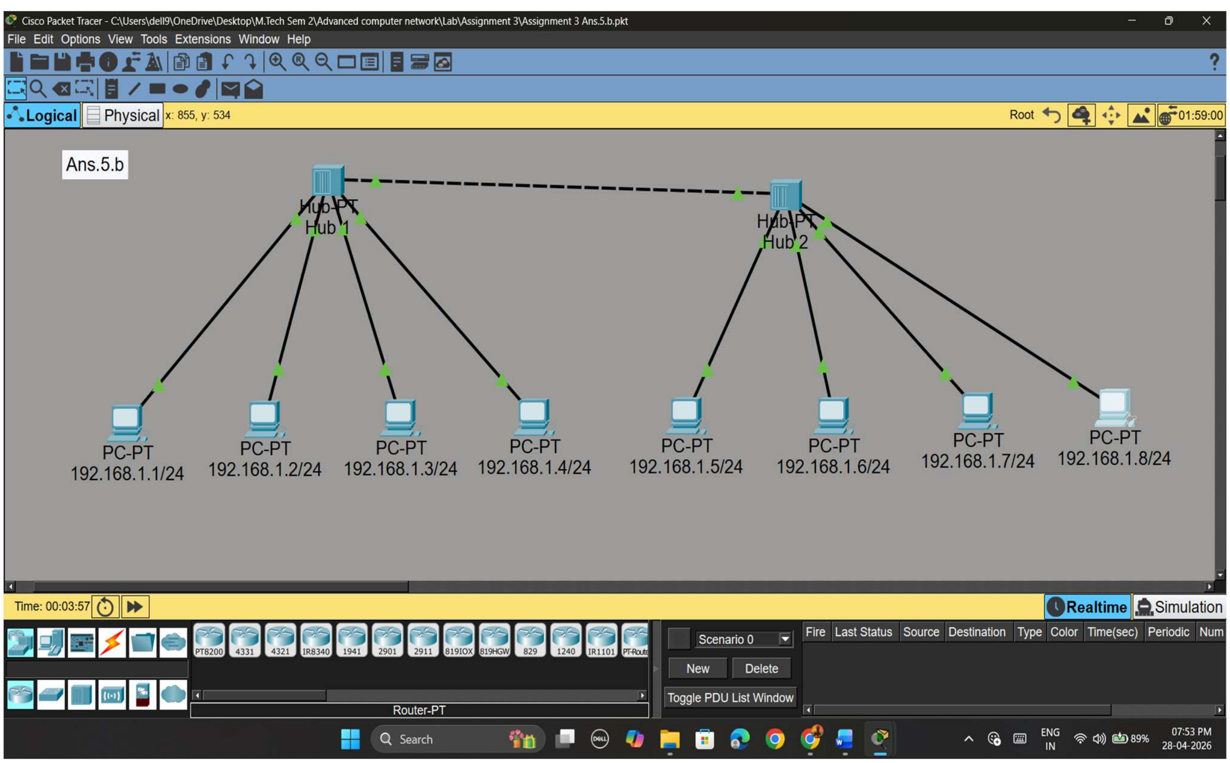
Ans.4


Ans.4
2960-24TT
Switch1
PC-PT
192.168.1.1/24
PC-PT
192.168.1.2/24

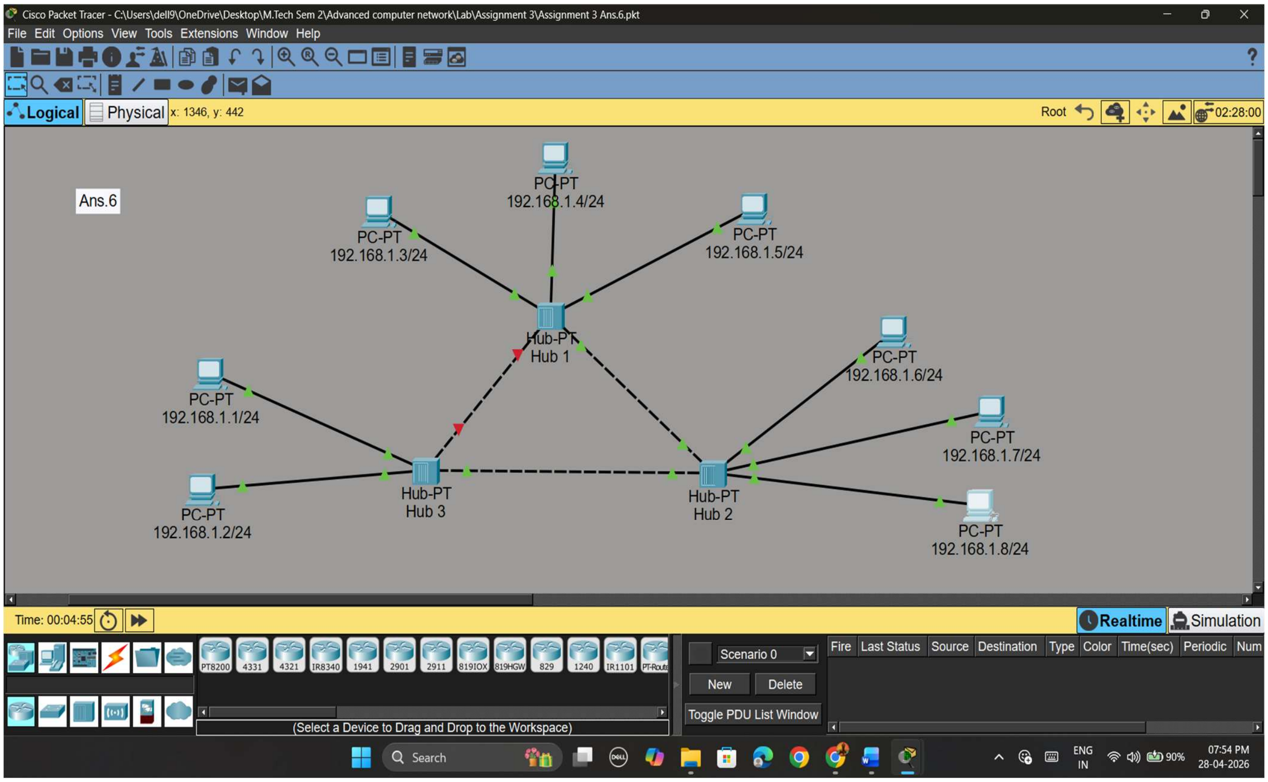
Ans.5(a)


Cisco Packet Tracer - C:\Users\dell9\OneDrive\Desktop\M.Tech Sem 2\Advanced computer network\Lab\Assignment 3\Assignment 3 Ans.5.a.pkt
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 1337, y: 459
Root
01:07:30
Ans.5.a
Hub-PT
Hub 1
PC-PT
192.168.1.1/24
PC-PT
192.168.1.2/24
PC-PT
192.168.1.3/24
PC-PT
192.168.1.4/24
PC-PT
192.168.1.5/24
PC-PT
192.168.1.6/24
PC-PT
192.168.1.7/24
PC-PT
192.168.1.8/24
Time: 00:02:14
Realtime
Simulation
Scenario 0
New
Delete
Toggle PDU List Window
Fire Last Status Source Destination Type Color Time(sec) Periodic Num
(Select a Device to Drag and Drop to the Workspace)
Search
07:51 PM
28-04-2026

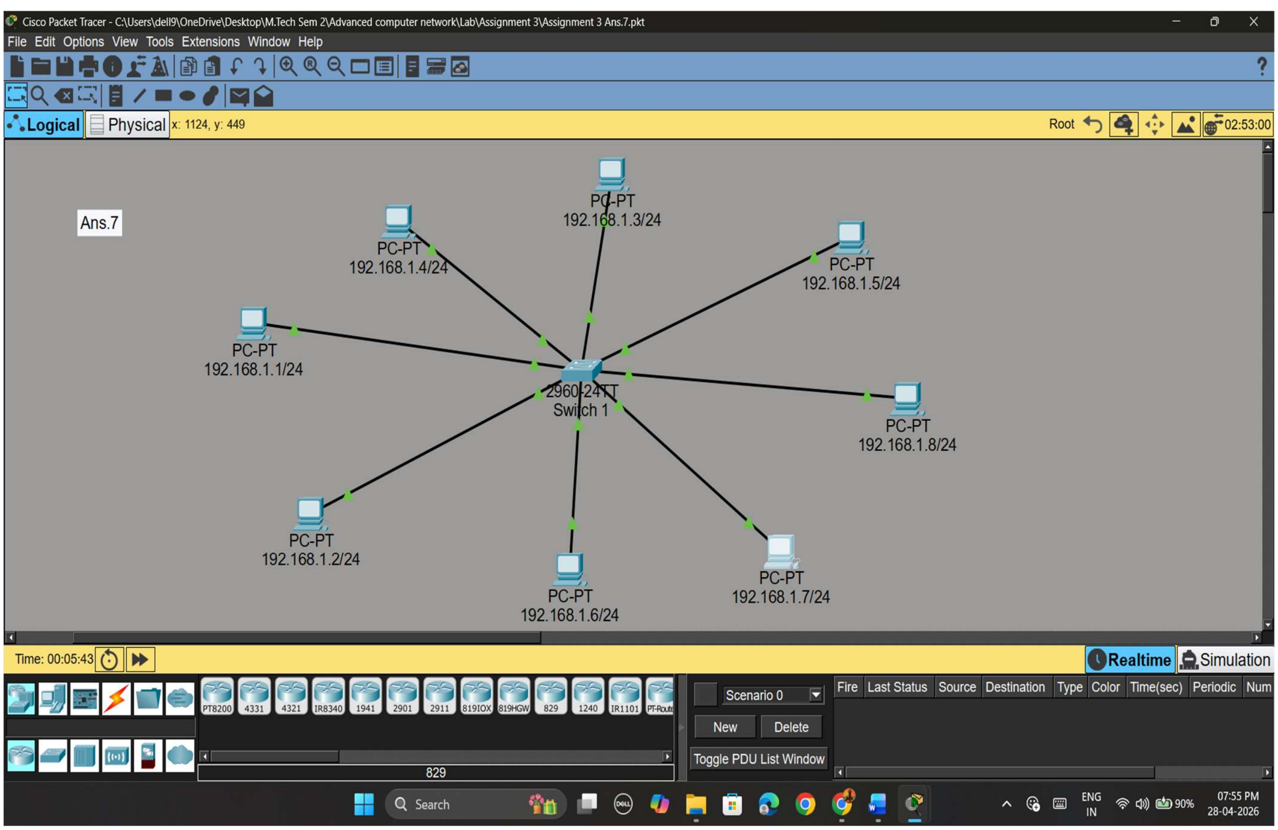
Ans.5(b)


Cisco Packet Tracer - C:\Users\dell9\OneDrive\Desktop\M.Tech Sem 2\Advanced computer network\Lab\Assignment 3\Assignment 3 Ans.5.b.pkt
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 855, y: 534
Root
01:59:00
Ans.5.b
Hub-PT
Hub 1
Hub-PT
Hub 2
PC-PT
192.168.1.1/24
PC-PT
192.168.1.2/24
PC-PT
192.168.1.3/24
PC-PT
192.168.1.4/24
PC-PT
192.168.1.5/24
PC-PT
192.168.1.6/24
PC-PT
192.168.1.7/24
PC-PT
192.168.1.8/24
Time: 00:03:57
Realtime
Simulation
Scenario 0
New
Delete
Toggle PDU List Window
Fire Last Status Source Destination Type Color Time(sec) Periodic Num
Router-PT
Search
07:53 PM
28-04-2026

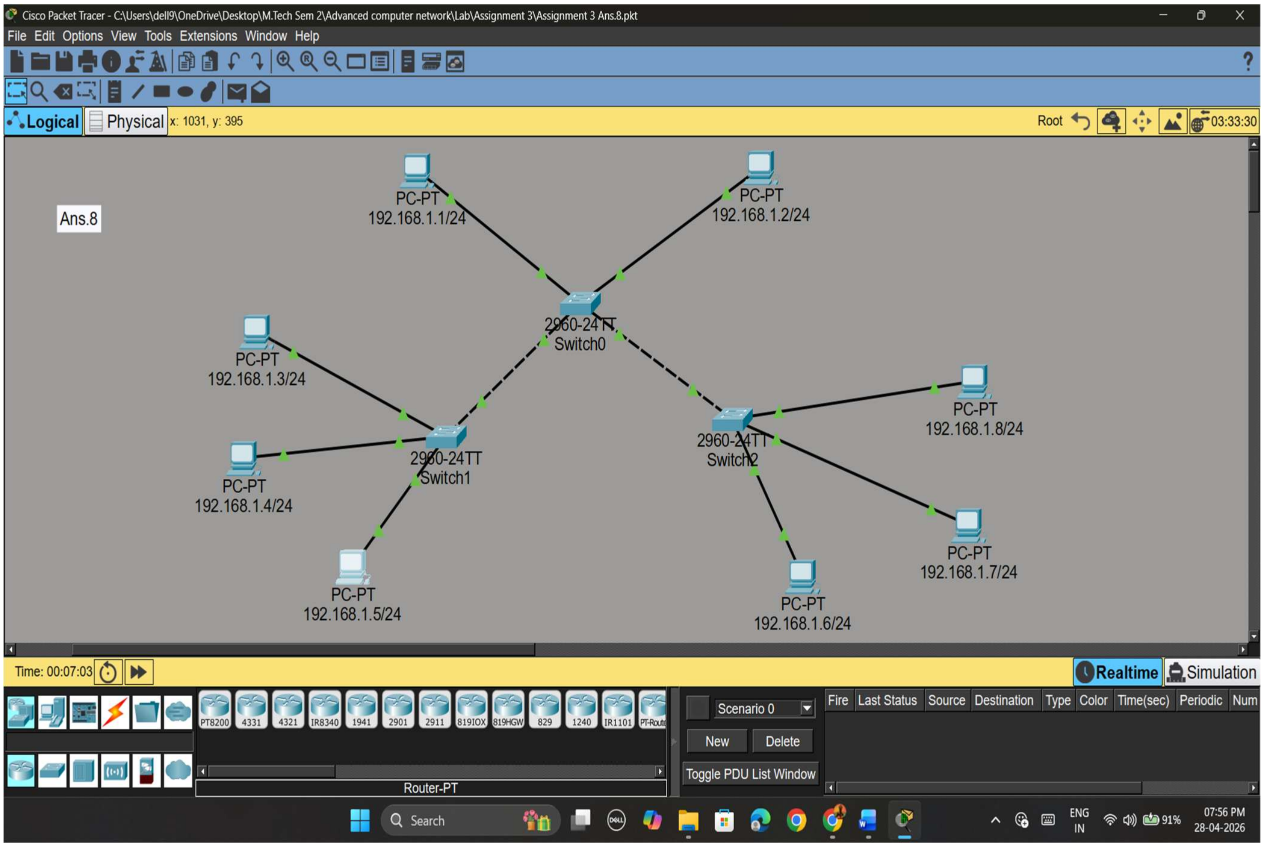
Ans.6


Cisco Packet Tracer - C:\Users\dell9\OneDrive\Desktop\M.Tech Sem 2\Advanced computer network\Lab\Assignment 3\Assignment 3 Ans.6.pkt
Ans.6
PC-PT
192.168.1.4/24
PC-PT
192.168.1.3/24
PC-PT
192.168.1.5/24
Hub-PT
Hub 1
PC-PT
192.168.1.6/24
PC-PT
192.168.1.1/24
PC-PT
192.168.1.7/24
PC-PT
192.168.1.2/24
Hub-PT
Hub 3
Hub-PT
Hub 2
PC-PT
192.168.1.8/24
Time: 00:04:55
Realtime
Simulation
(Select a Device to Drag and Drop to the Workspace)

Ans.7

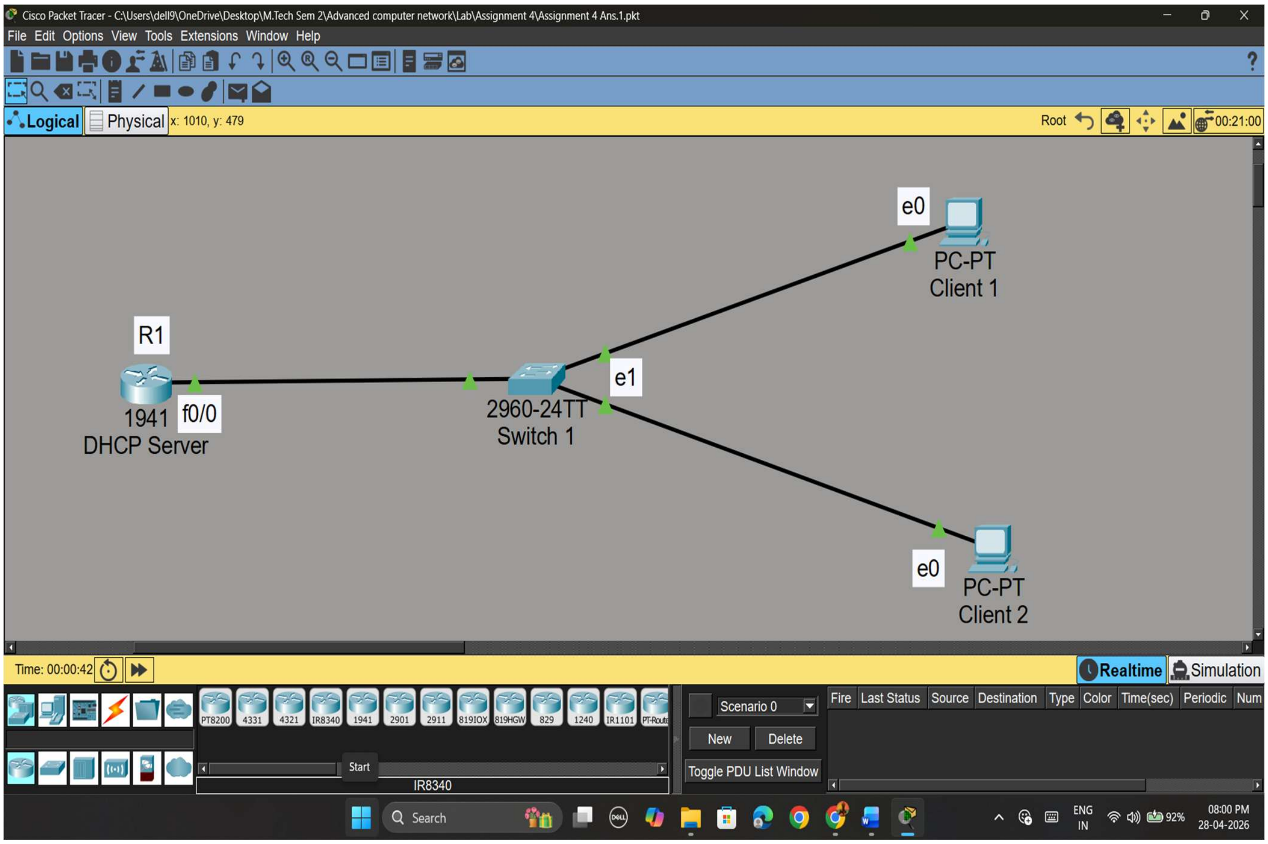

Cisco Packet Tracer - C:\Users\dell9\OneDrive\Desktop\M.Tech Sem 2\Advanced computer network\Lab\Assignment 3\Assignment 3 Ans.7.pkt
Ans.7
PC-PT
192.168.1.3/24
PC-PT
192.168.1.4/24
PC-PT
192.168.1.5/24
PC-PT
192.168.1.1/24
2960-24TT
Switch 1
PC-PT
192.168.1.8/24
PC-PT
192.168.1.2/24
PC-PT
192.168.1.6/24
PC-PT
192.168.1.7/24
Time: 00:05:43
Realtime
Simulation

Ans.8



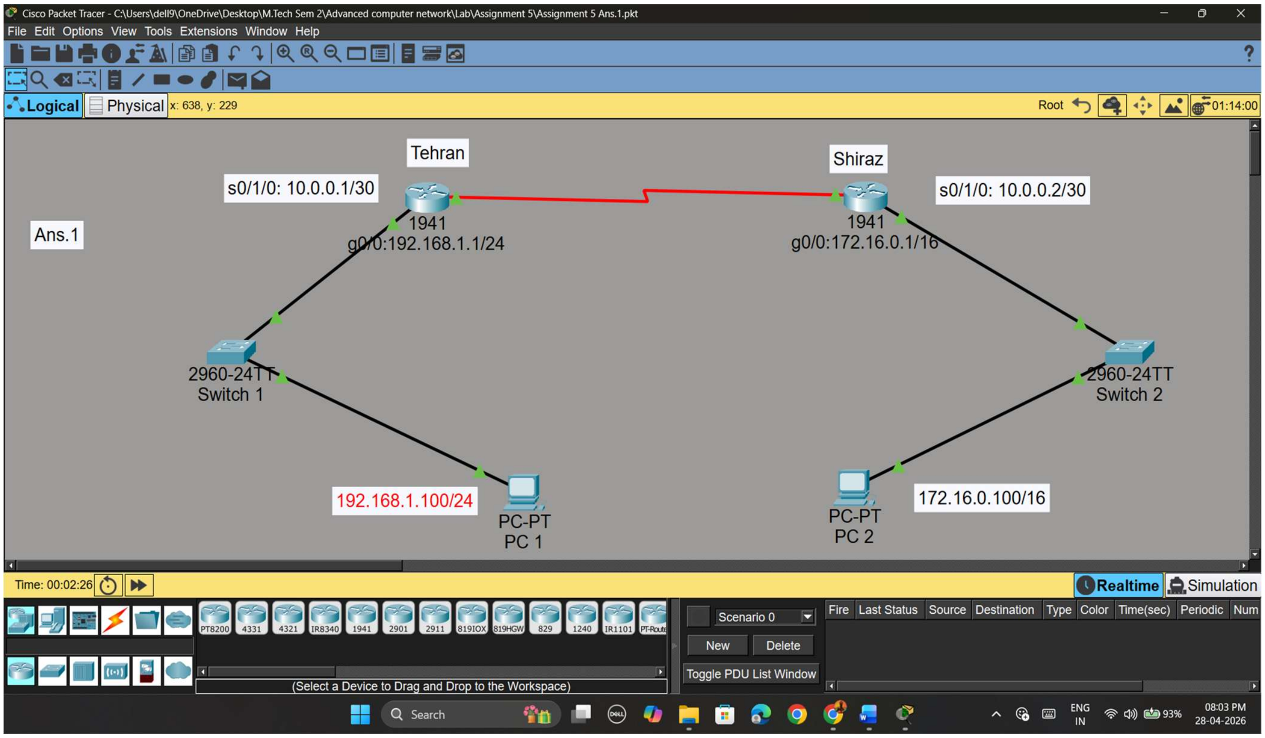
Cisco Packet Tracer - C:\Users\dell9\OneDrive\Desktop\M.Tech Sem 2\Advanced computer network\Lab\Assignment 3\Assignment 3 Ans.8.pkt
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 1031, y: 395
Root
03:33:30
Ans.8
PC-PT
192.168.1.1/24
PC-PT
192.168.1.2/24
2960-24TT
Switch0
PC-PT
192.168.1.3/24
PC-PT
192.168.1.4/24
2960-24TT
Switch1
PC-PT
192.168.1.5/24
2960-24TT
Switch2
PC-PT
192.168.1.8/24
PC-PT
192.168.1.7/24
PC-PT
192.168.1.6/24
Time: 00:07:03
Realtime
Simulation
Scenario 0
New
Delete
Toggle PDU List Window
Fire
Last Status
Source
Destination
Type
Color
Time(sec)
Periodic
Num
Router-PT
Search
07:56 PM
28-04-2026

ASSIGNMENT 4

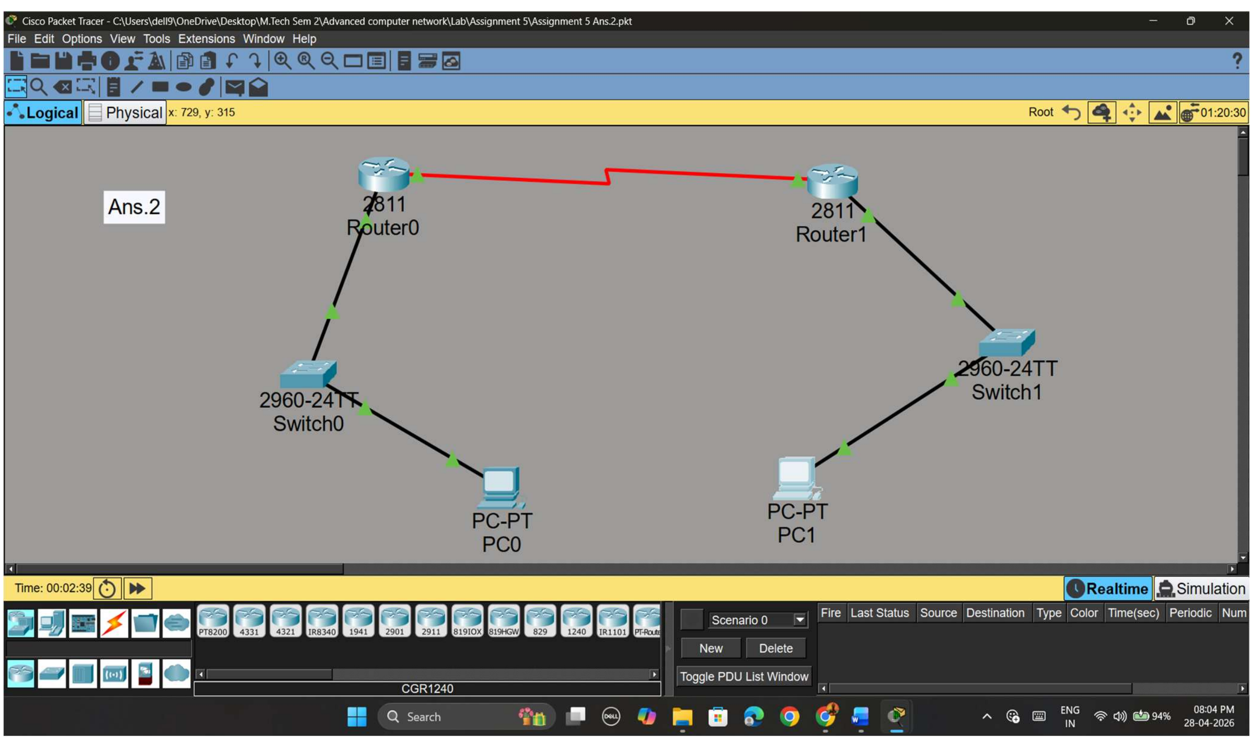


ASSIGNMENT 5

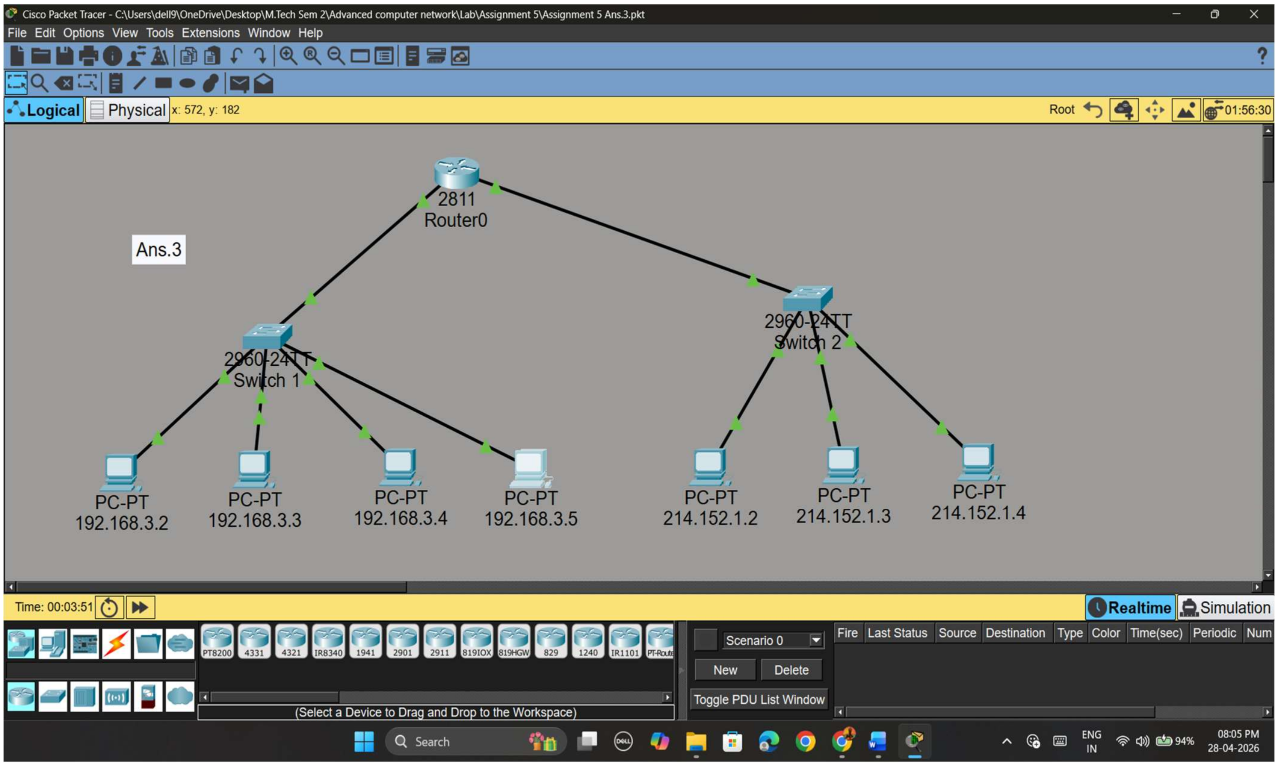
Ans.1



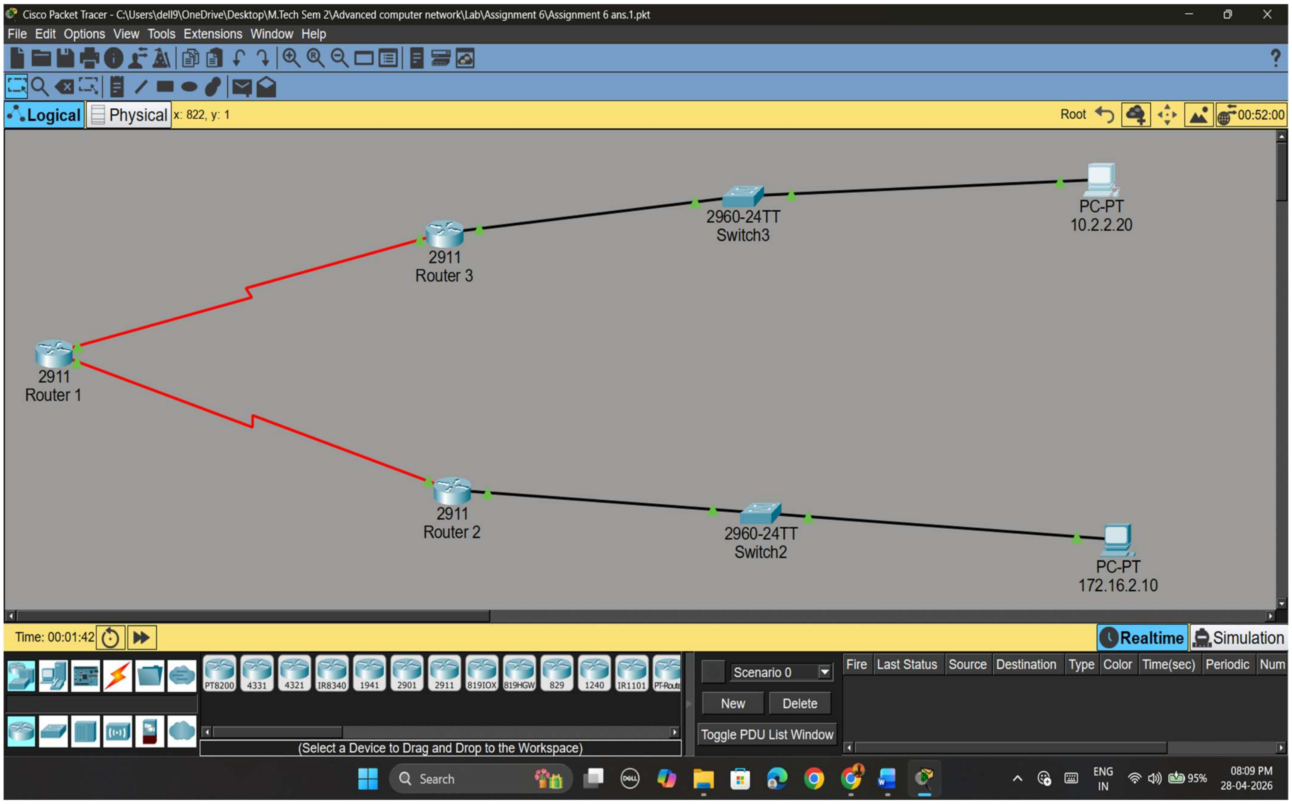
Ans.2



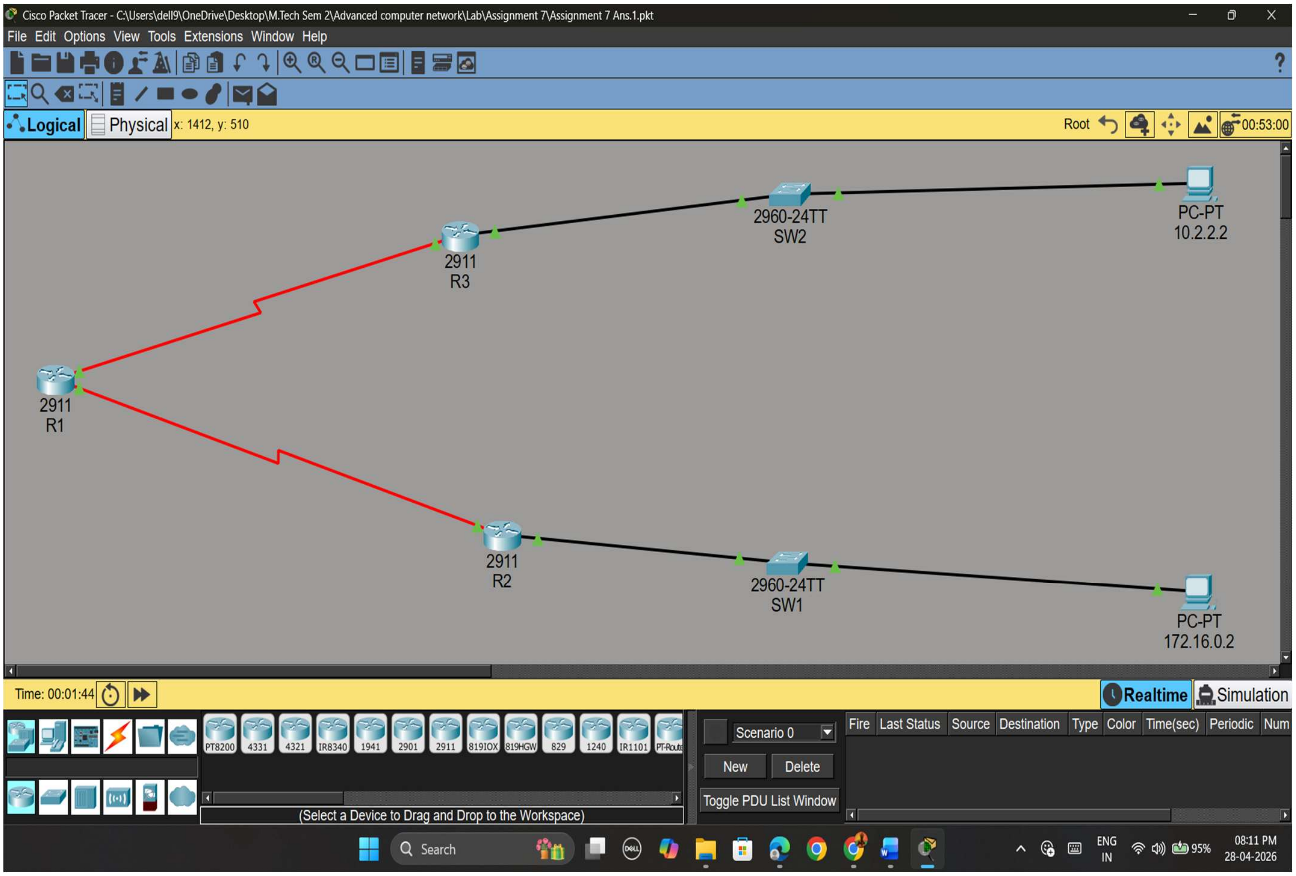
Ans.3


Cisco Packet Tracer - C:\Users\dell9\OneDrive\Desktop\M.Tech Sem 2\Advanced computer network\Lab\Assignment 5\Assignment 5 Ans.3.pkt
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 572, y: 182
Root
01:56:30
Ans.3
2811
Router0
2960-24TT
Switch 1
2960-24TT
Switch 2
PC-PT
192.168.3.2
PC-PT
192.168.3.3
PC-PT
192.168.3.4
PC-PT
192.168.3.5
PC-PT
214.152.1.2
PC-PT
214.152.1.3
PC-PT
214.152.1.4
Time: 00:03:51
Realtime
Simulation
Scenario 0
New
Delete
Toggle PDU List Window
Fire Last Status Source Destination Type Color Time(sec) Periodic Num
(Select a Device to Drag and Drop to the Workspace)
Search
ENG
IN
94%
08:05 PM
28-04-2026

ASSIGNMENT 6



ASSIGNMENT 7



Additional Analytical Task (NEW)

Fill the table:

Router	Destination	Path	Cost
R1	10.2.2.0	R1→R3	64

ASSIGNMENT 8